Storyline

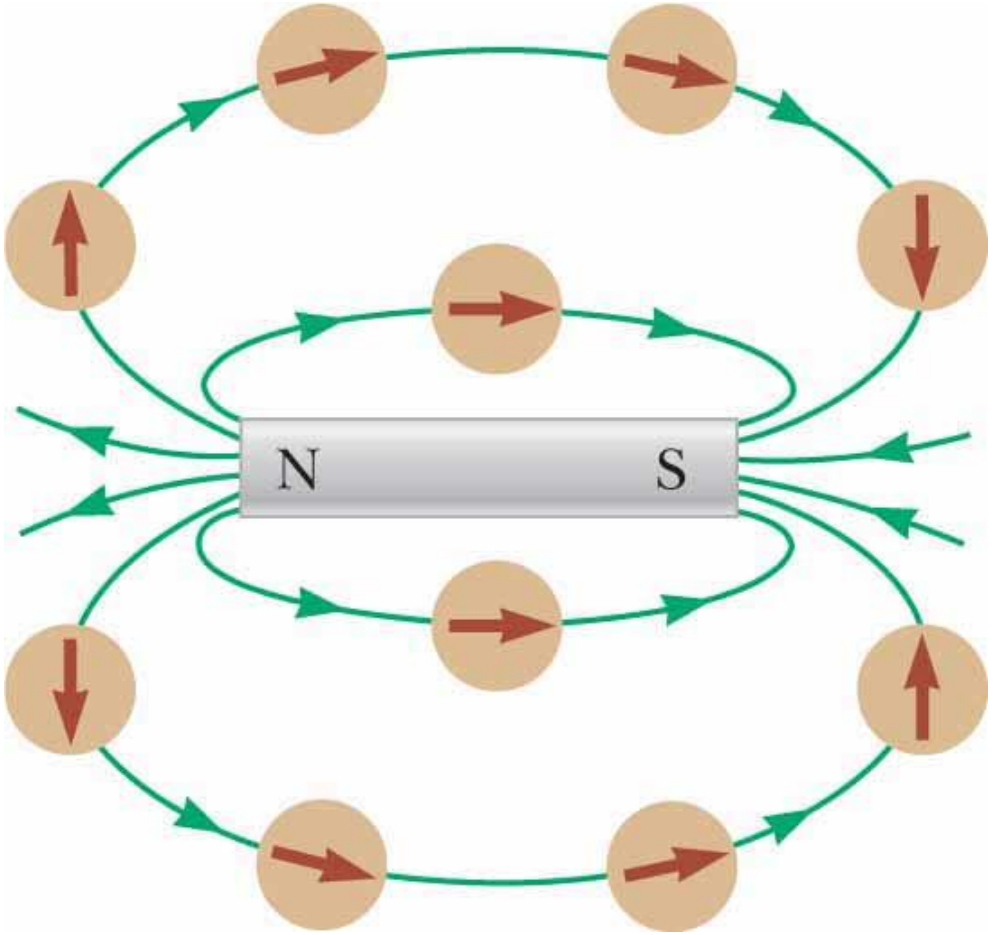
Chapter 28: Magnetic Fields



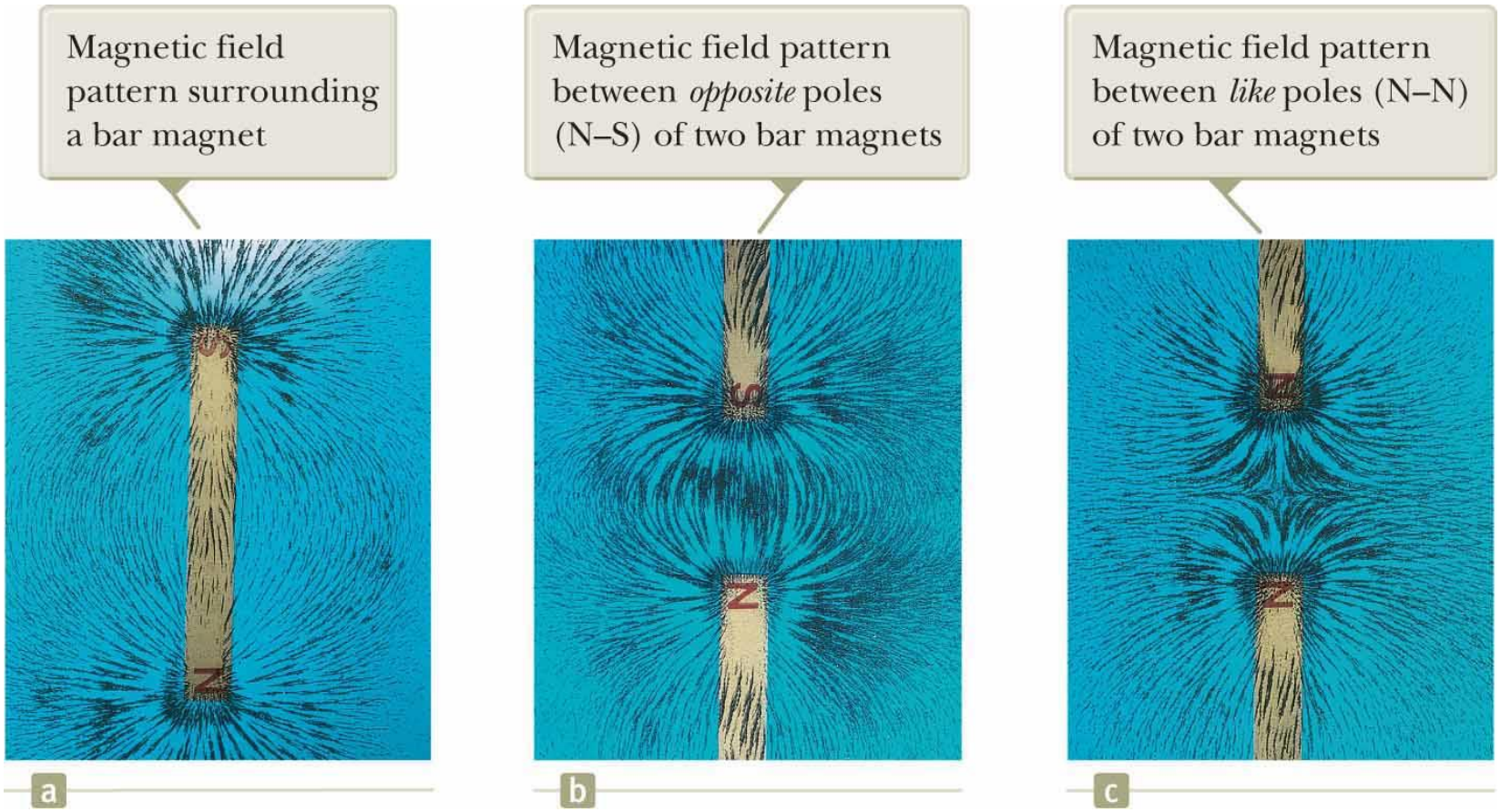
Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

BROOKS/COLE
CENGAGE Learning

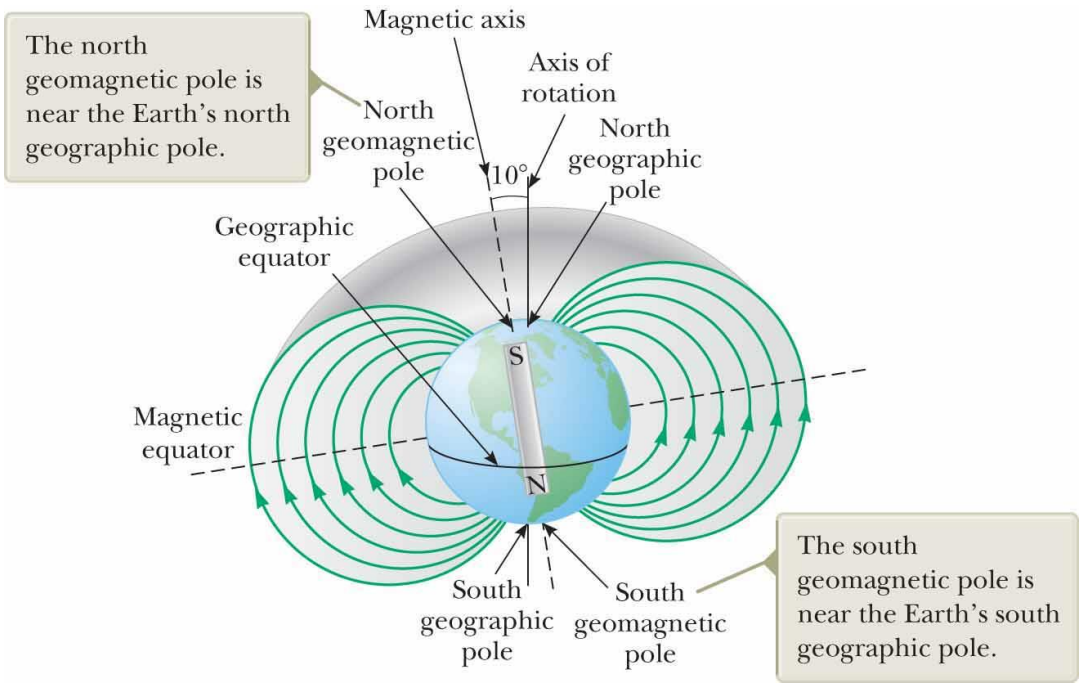
Analysis Model: Particle in a Field (Magnetic)

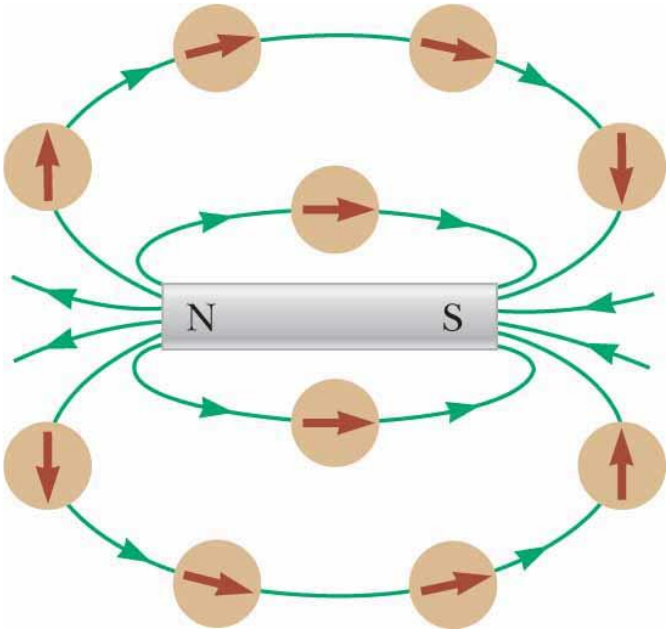


Magnetic Field Patterns of Bar Magnets



Earth's Magnetic Field



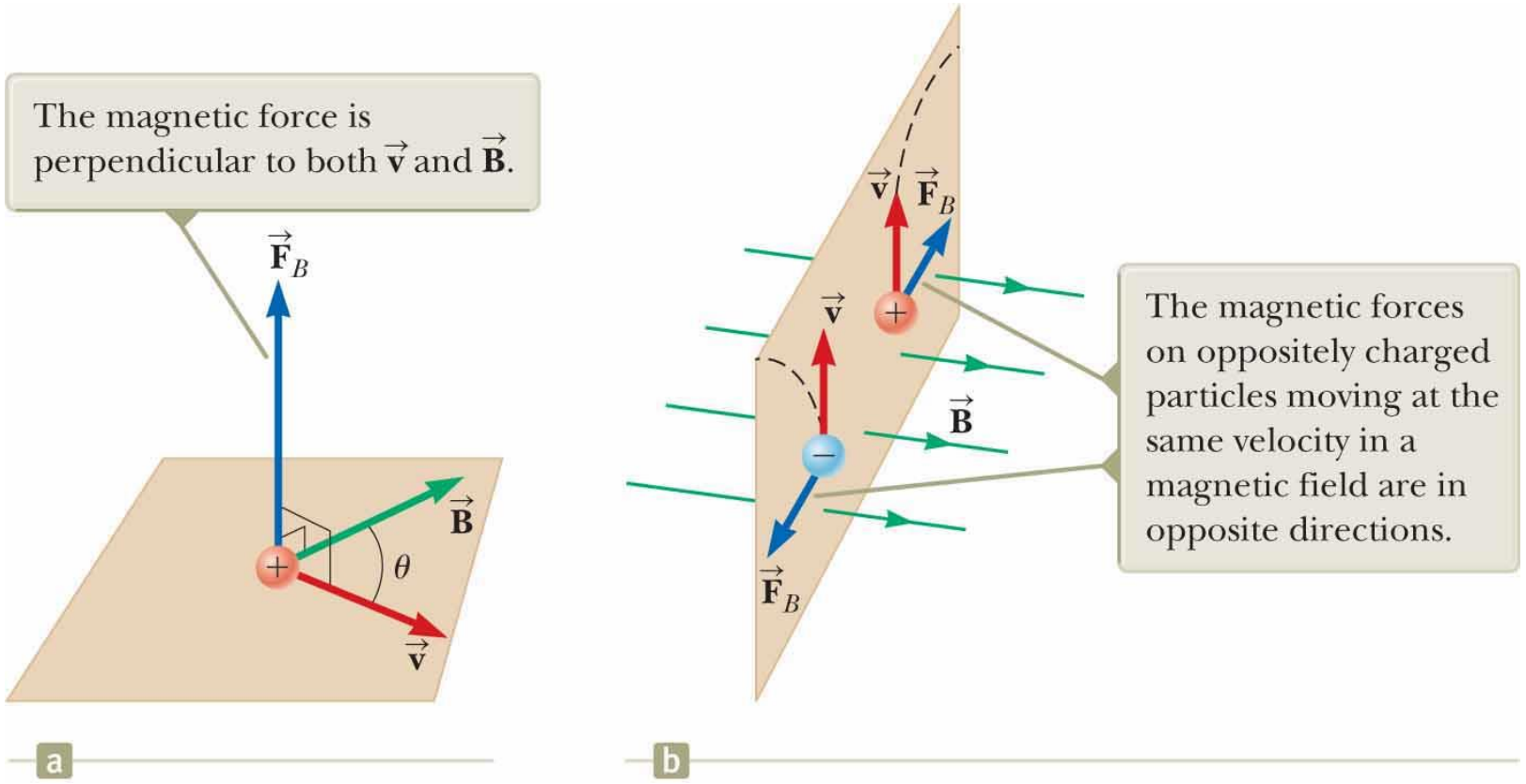


Analysis Model: Particle in a Field (Magnetic)

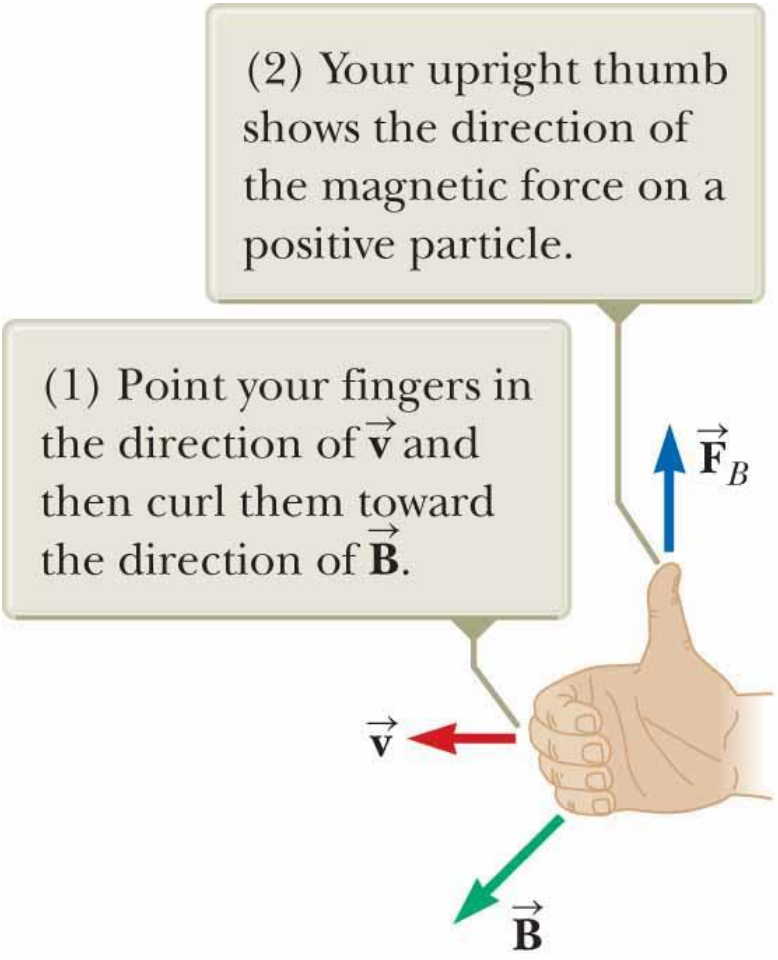
- Magnetic force proportional to charge q of particle.
- Magnetic force on negative charge directed opposite to force on positive charge moving in same direction.
- Magnetic force proportional to magnitude of magnetic field vector **B**.

Analysis Model: Particle in a Field (Magnetic)

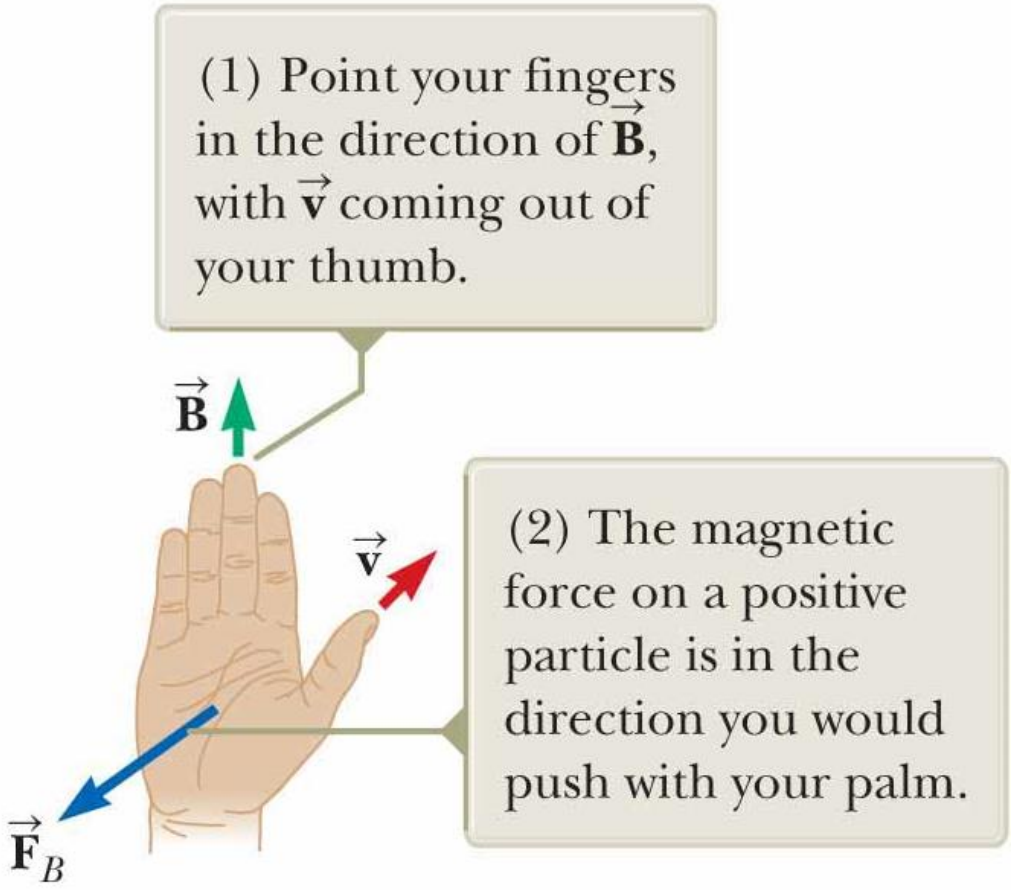
$$\vec{\mathbf{F}}_B = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$



Right-Hand Rule



Alternative Right-Hand Rule



Analysis Model: Particle in a Field (Magnetic)

$$\vec{\mathbf{F}}_B = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

$$F_B = |q| vB \sin\theta$$

$$1\ \mathrm{T} = 1\frac{\mathrm{N}}{\mathrm{C}\cdot\mathrm{m/s}} \qquad 1\ \mathrm{T} = 1\frac{\mathrm{N}}{\mathrm{A}\cdot\mathrm{m}}$$

$$1\ \mathrm{T} = 10^4\ \mathrm{G}$$

TABLE 28.1 Some Approximate Magnetic Field Magnitudes

Source of Field	Field Magnitude (T)
Strong superconducting laboratory magnet	30
Strong conventional laboratory magnet	2
Medical MRI unit	1.5
Bar magnet	10^{-2}
Surface of the Sun	10^{-2}
Surface of the Earth	5×10^{-5}
Inside human brain (due to nerve impulses)	10^{-13}

Quick Quiz 28.1

An electron moves in the plane of this screen toward the top of the screen. A magnetic field is also in the plane of the screen and directed toward the right. What is the direction of the magnetic force on the electron?

(a) toward the top of the screen
(b) toward the bottom of the screen
(c) toward the left edge of the screen
(d) toward the right edge of the screen
(e) out of the screen
(f) into the screen

Quick Quiz 28.1

An electron moves in the plane of this screen toward the top of the screen. A magnetic field is also in the plane of the screen and directed toward the right. What is the direction of the magnetic force on the electron?

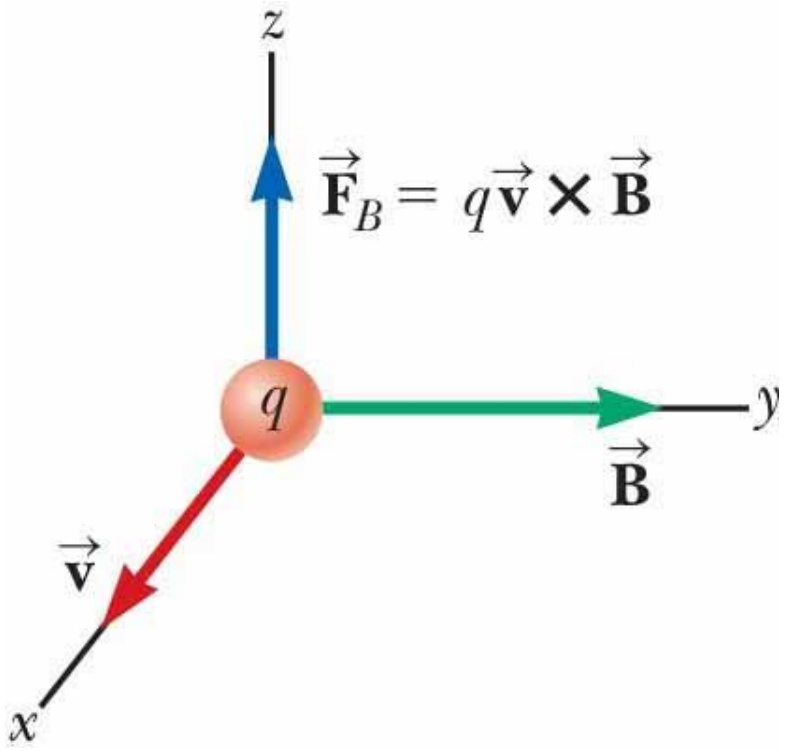
(a) toward the top of the screen
(b) toward the bottom of the screen
(c) toward the left edge of the screen
(d) toward the right edge of the screen
(e) out of the screen
(f) into the screen

Comparison of Electric and Magnetic Particle in Field Models

• Electric force vector along direction of electric field	• Magnetic force vector perpendicular to magnetic field
• Electric force acts on charged particle regardless of whether particle is moving	• Magnetic force acts on charged particle only when particle is in motion
• Electric force does work in displacing a charged particle	• Magnetic force associated with steady magnetic field does no work when a particle is displaced $\rightarrow$ force perpendicular to displacement of its point of application

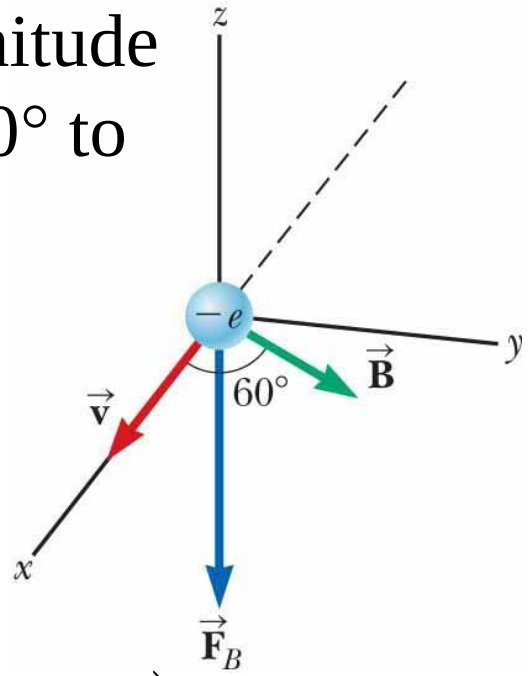
Analysis Model: Particle in a Field (Magnetic)

$$\vec{\mathbf{F}}_B = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$



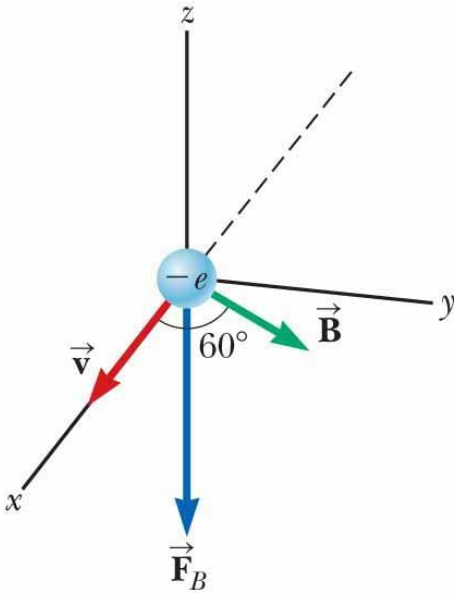
Example 28.1:
An Electron Moving in a Magnetic Field

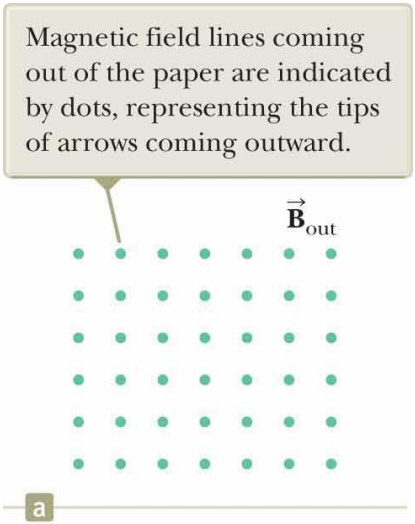
An electron moves through space as a cosmic ray with a speed of 8.0×10^6 m/s along the *x* axis. At its location, the magnetic field of the Earth has a magnitude 0.050 mT, and is directed at an angle of 60° to the *x* axis, lying in the *xy* plane.
Calculate the magnetic force on the electron.

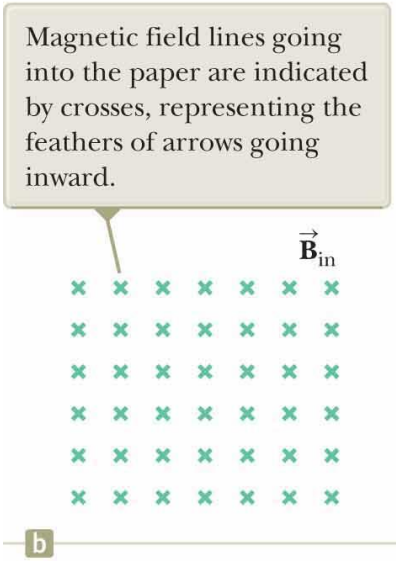


$$F_B = |q| vB \sin\theta$$
$$= \left(1.6\times10^{-19}\ \text{C}\right)\left(8.0\times10^{6}\ \text{m/s}\right)\left(5.0\times10^{-5}\ \text{T}\right)\left(\sin 60°\right)$$
$$= 5.5\times10^{-17}\ \text{N}$$

Motion of a Charged Particle in a Uniform Magnetic Field







Motion of a Charged Particle in a Uniform Magnetic Field

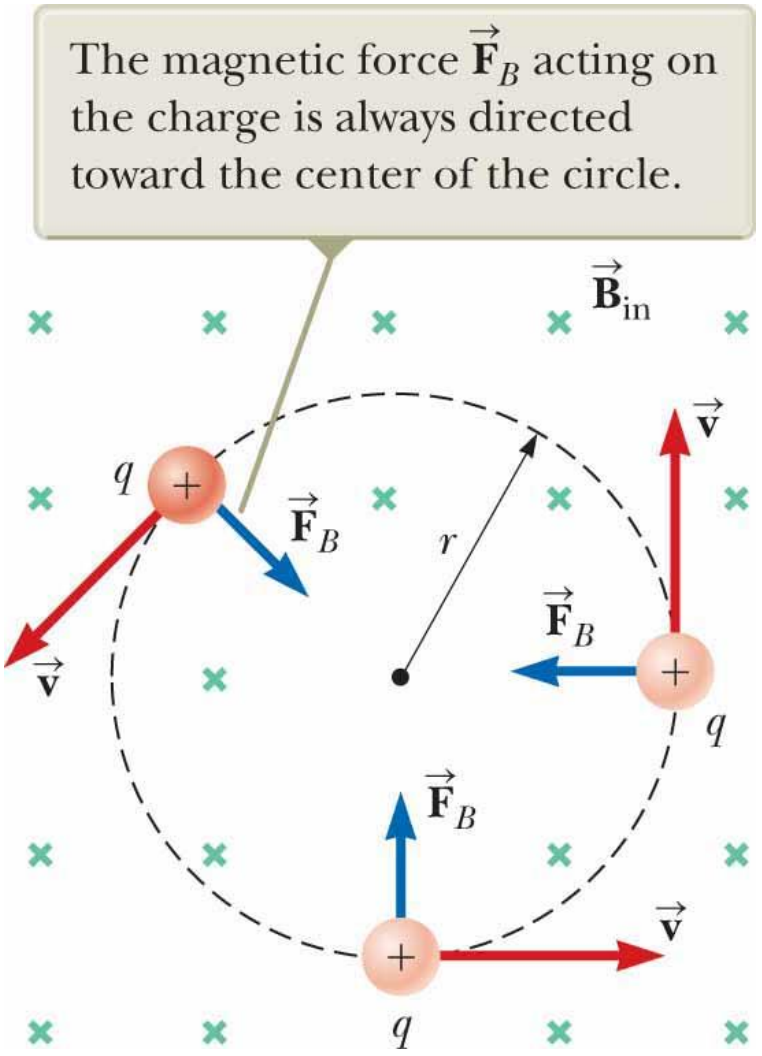
$$\sum F = F_B = ma$$

$$F_B = qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB}$$

$$\omega = \frac{v}{r} = \frac{qB}{m}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$



Quick Quiz 28.2 Part I

A charged particle is moving perpendicular to a magnetic field in a circle with a radius *r*.

An identical particle enters the field, with $\vec{\mathbf{v}}$ perpendicular to $\vec{\mathbf{B}}$, but with a higher speed than the first particle. Compared with the radius of the circle for the first particle, is the radius of the circular path for the second particle

(a) smaller,
(b) larger, or
(c) equal in size?

Quick Quiz 28.2 Part I

A charged particle is moving perpendicular to a magnetic field in a circle with a radius *r*.

An identical particle enters the field, with $\vec{\mathbf{v}}$ perpendicular to $\vec{\mathbf{B}}$, but with a higher speed than the first particle. Compared with the radius of the circle for the first particle, is the radius of the circular path for the second particle

(a) smaller,

(b) larger, or

(c) equal in size?

Quick Quiz 28.2 Part II

A charged particle is moving perpendicular to a magnetic field in a circle with a radius *r*.
The magnitude of the magnetic field is increased.
Compare the radius of the new circular path of the first particle with the radius of its initial path.

(a) smaller,
(b) larger, or
(c) equal in size

Quick Quiz 28.2 Part II

A charged particle is moving perpendicular to a magnetic field in a circle with a radius *r*.
The magnitude of the magnetic field is increased.
Compare the radius of the new circular path of the first particle with the radius of its initial path.

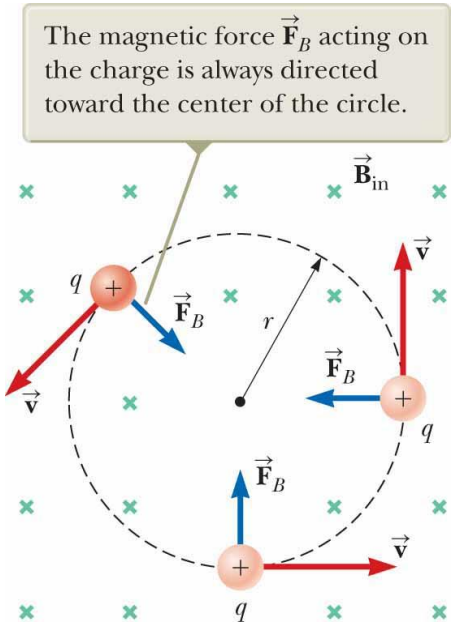
(a) smaller,
(b) larger, or
(c) equal in size

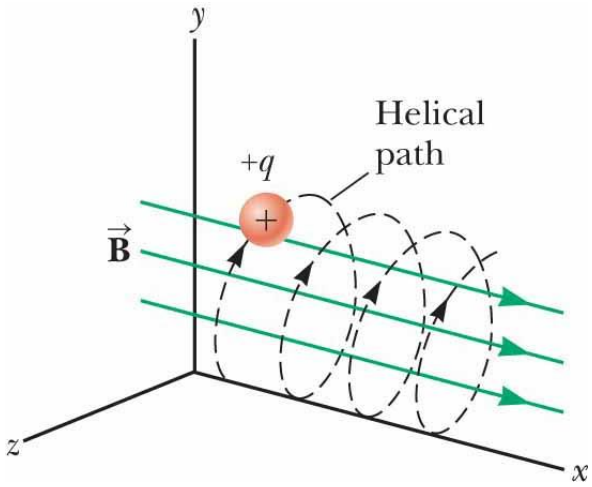
Motion of a Charged Particle in a Uniform Magnetic Field

$$r = \frac{mv}{qB} \qquad \omega = \frac{v}{r} = \frac{qB}{m}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

$$v \rightarrow v_{\perp} = \sqrt{v_y^{\,2} + v_z^{\,2}}$$





Example 28.2: A Proton Moving Perpendicular to a Uniform Magnetic Field

A proton is moving in a circular orbit of radius 14 cm in a uniform 0.35-T magnetic field perpendicular to the velocity of the proton. Find the speed of the proton.

$$v = \frac{vBr}{m_p}$$

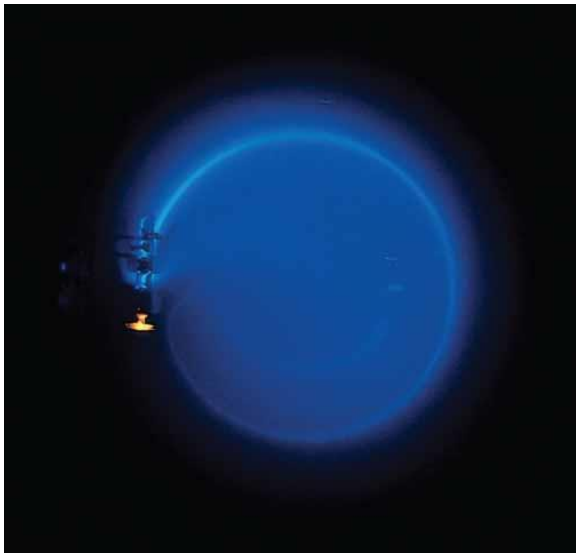
$$v = \frac{\left(1.60\times10^{-19}\ \text{C}\right)\left(0.35\ \text{T}\right)\left(0.14\ \text{m}\right)}{1.67\times10^{-27}\ \text{kg}} = \boxed{4.7\times10^{6}\ \text{m/s}}$$

Example 28.2: A Proton Moving Perpendicular to a Uniform Magnetic Field

What if an electron, rather than a proton, moves in a direction perpendicular to the same magnetic field with this same speed? Will the radius of its orbit be different?

Example 28.3:
Bending an Electron Beam

In an experiment designed to measure the magnitude of a uniform magnetic field, electrons are accelerated from rest through a potential difference of 350 V and then enter a uniform magnetic field that is perpendicular to the velocity vector of the electrons. The electrons travel along a curved path because of the magnetic force exerted on them, and the radius of the path is measured to be 7.5 cm. (Such a curved beam of electrons is shown in the figure.)



(A)What is the magnitude of the magnetic field?

Example 28.3:
Bending an Electron Beam

$$\Delta K + \Delta U_E = 0 \quad \Rightarrow \left(\frac{1}{2} m_e v^2 - 0\right) + \left(q\Delta V\right) = 0$$

$$\Rightarrow v = \sqrt{\frac{-2q\Delta V}{m_e}}$$

$$v = \sqrt{\frac{-2\left(-1.60\times10^{-19}\ \text{C}\right)\left(350\ \text{V}\right)}{9.11\times10^{-31}\ \text{kg}}} = 1.11\times10^{7}\ \text{m/s}$$

$$B = \frac{m_e v}{er} = \frac{\left(9.11\times10^{-31}\ \text{kg}\right)\left(1.11\times10^{7}\ \text{m/s}\right)}{\left(1.60\times10^{-19}\ \text{C}\right)\left(0.075\ \text{m}\right)} = \boxed{8.4\times10^{-4}\ \text{T}}$$

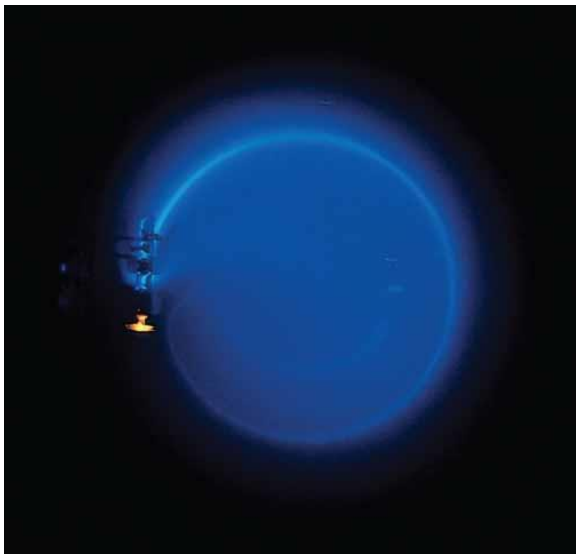
Example 28.3: Bending an Electron Beam

(B) What is the angular speed of the electrons?

$$\omega = \frac{v}{r} = \frac{1.11\times10^{7}\ \text{m/s}}{0.075\ \text{m}} = \boxed{1.5\times10^{8}\ \text{rad/s}}$$

$$\omega = \left(1.5\times10^{8}\ \text{rad/s}\right)\left(\frac{1\ \text{rev}}{2\pi\ \text{rad}}\right)$$

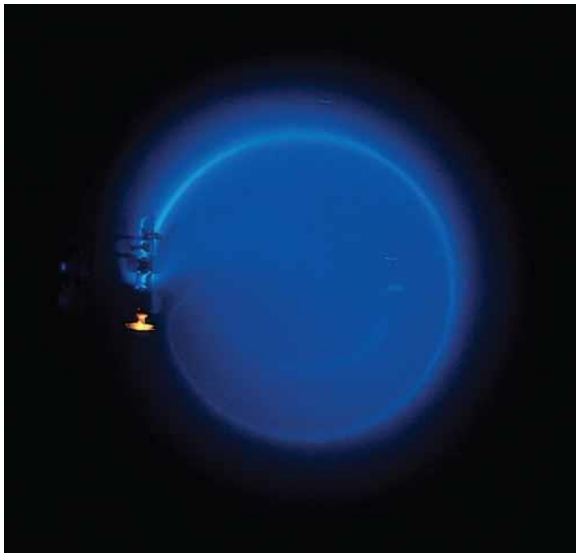
$$= 2.4\times10^{7}\ \text{rev/s}$$



Example 28.3:
Bending an Electron Beam

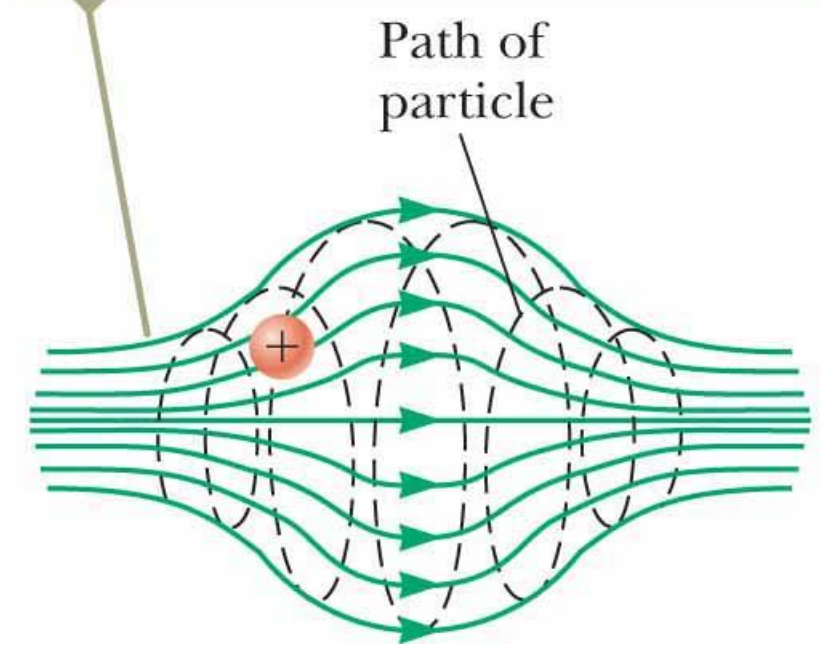
What if a sudden voltage surge causes the accelerating voltage to increase to 400 V? How does that affect the angular speed of the electrons, assuming the magnetic field remains constant?

$$\omega = \frac{qB}{m}$$

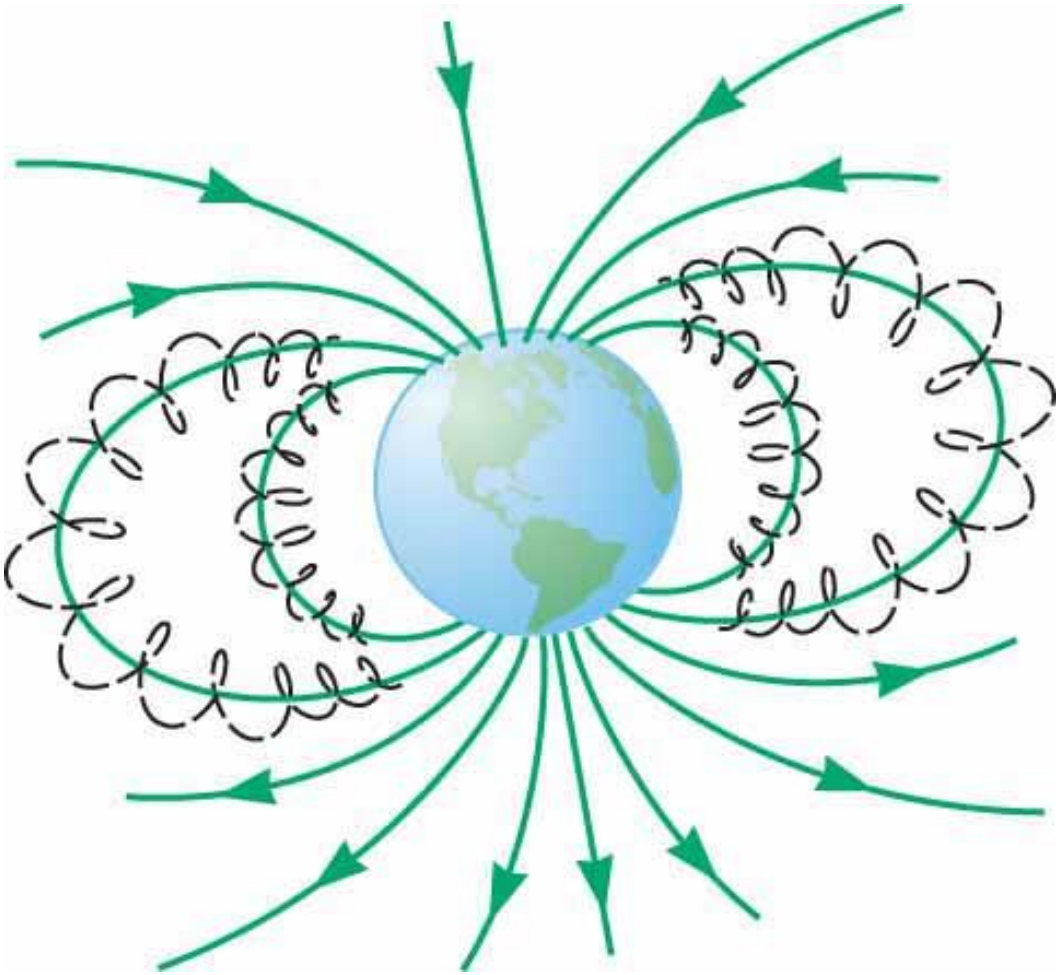


Magnetic Bottle

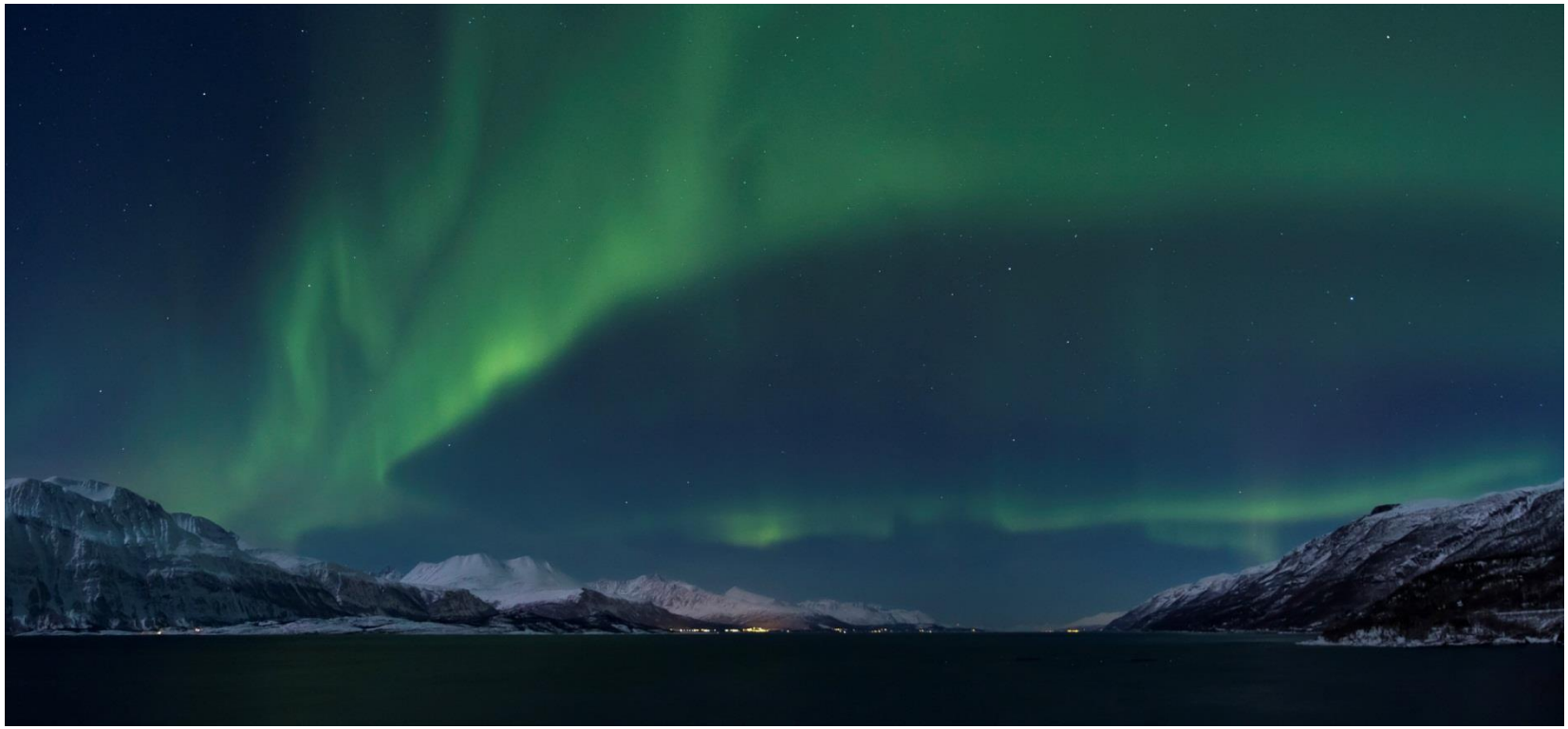
The magnetic force exerted on the particle near either end of the bottle has a component that causes the particle to spiral back toward the center.



Van Allen Radiation Belts



Auroras

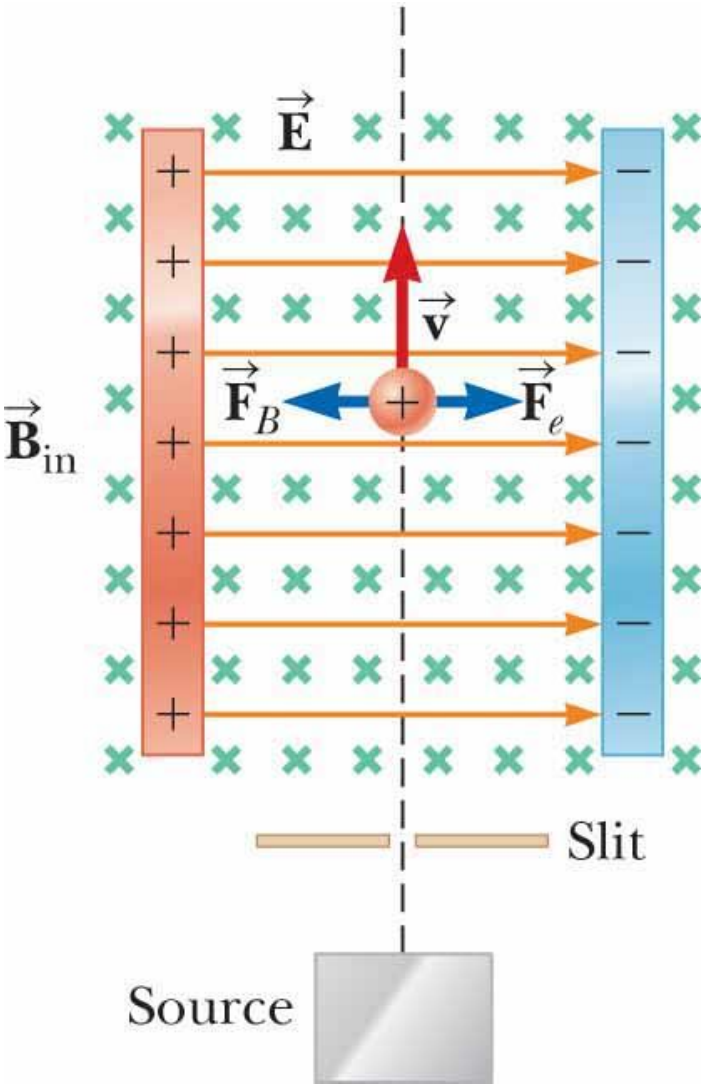


Velocity Selector

$$\vec{\mathbf{F}} = q\vec{\mathbf{E}} + q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

$$qE = qvB$$

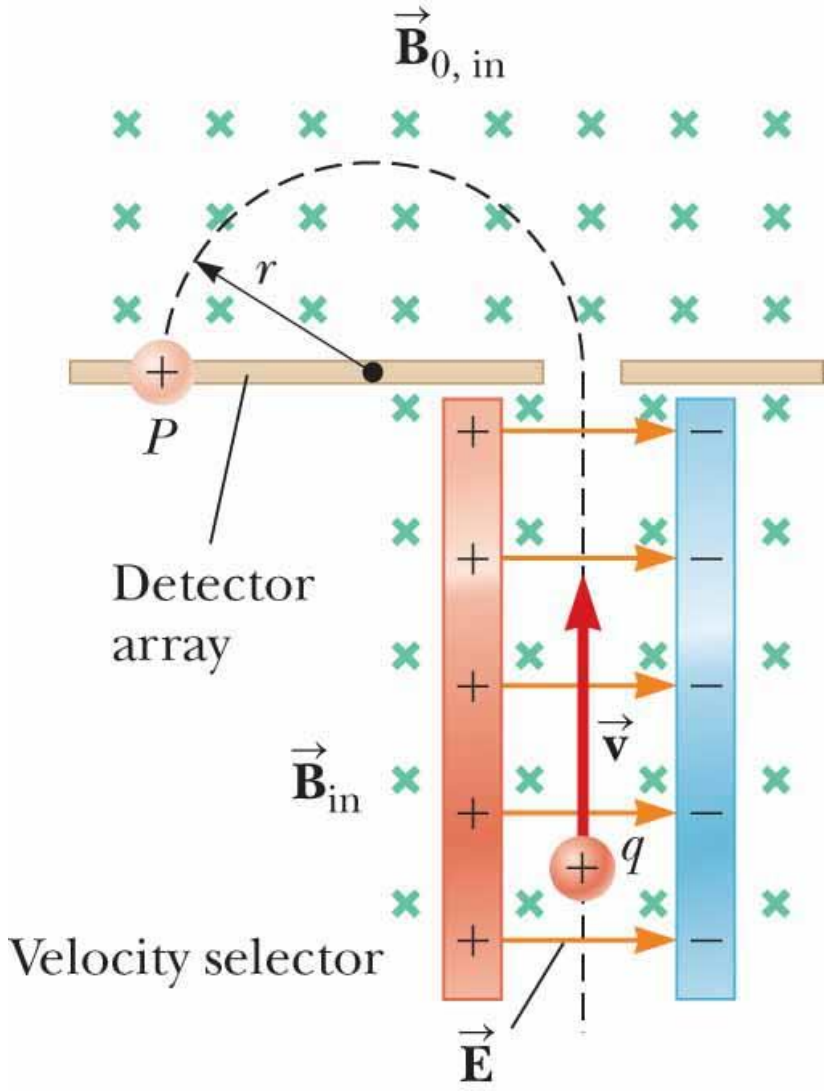
$$v = \frac{E}{B}$$



The Mass Spectrometer

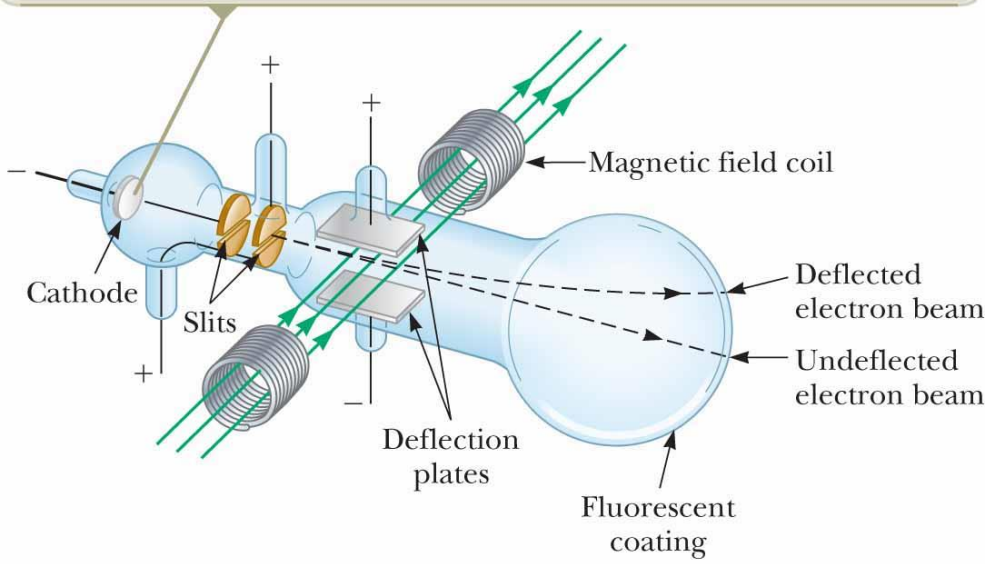
$$\frac{m}{q} = \frac{rB_0}{v}$$

$$\frac{m}{q} = \frac{rB_0 B}{E}$$



Measuring e/m_e

Electrons are accelerated from the cathode, pass through two slits, and are deflected by both an electric field (formed by the charged deflection plates) and a magnetic field (directed perpendicular to the electric field). The beam of electrons then strikes a fluorescent screen.



a

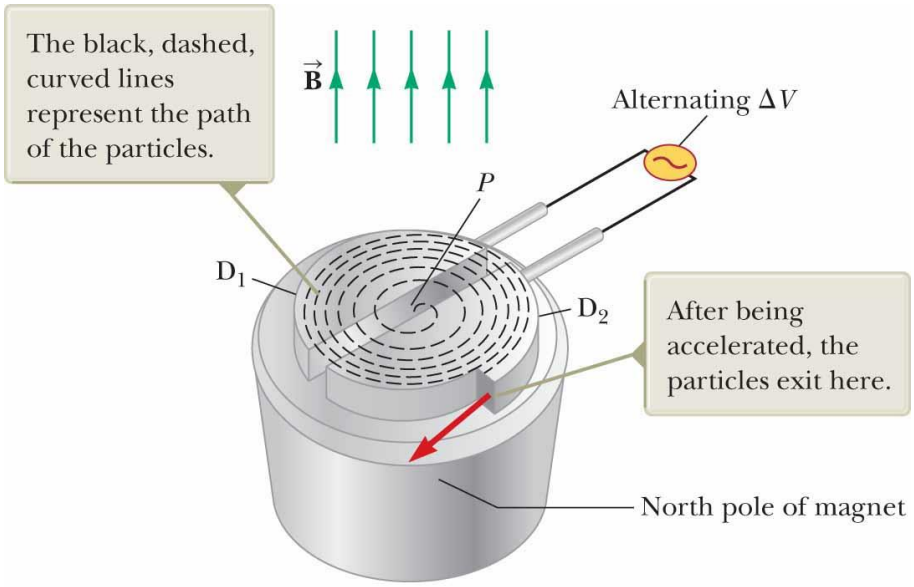


b

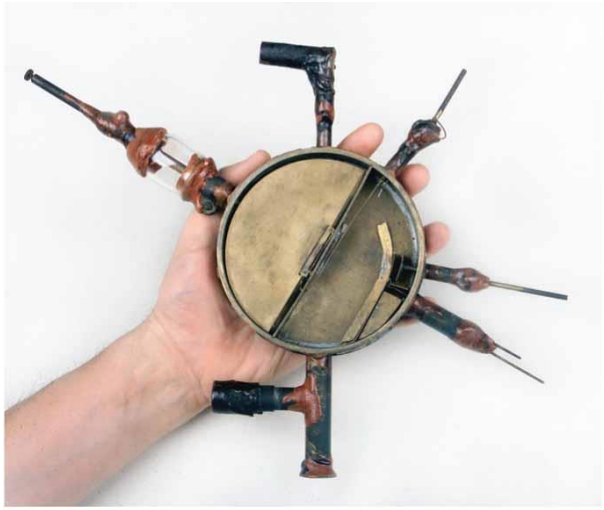
J. J. Thomson and
Frank Baldwin Jewett

The Cyclotron

$$T = \frac{2\pi m}{qB} \qquad v = \frac{qBR}{m} \Rightarrow K = \frac{1}{2}mv^2 = \frac{q^2B^2R^2}{2m}$$

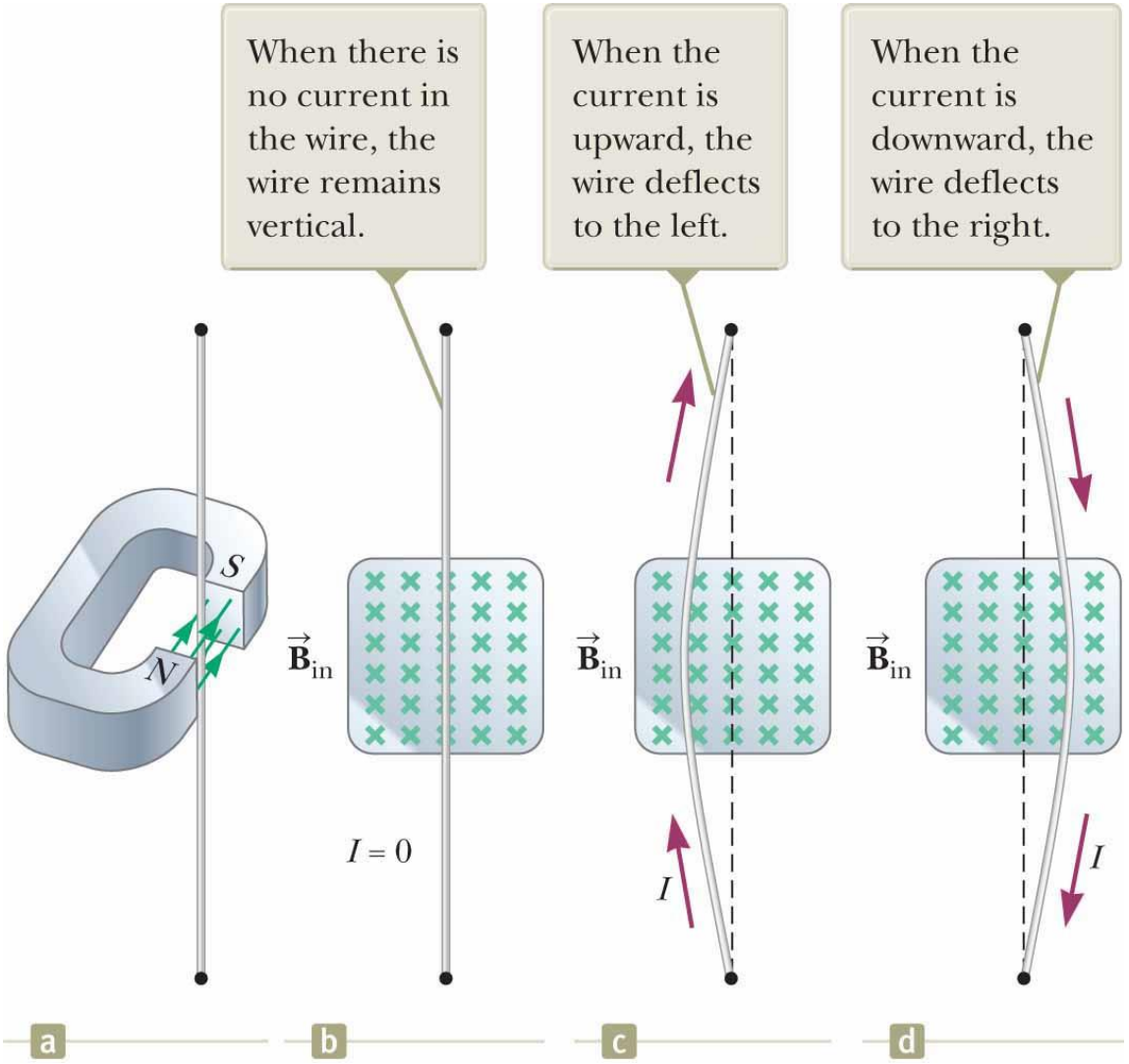


a



b

Magnetic Force Acting on a Current-Carrying Conductor



Magnetic Force Acting on a Current-Carrying Conductor

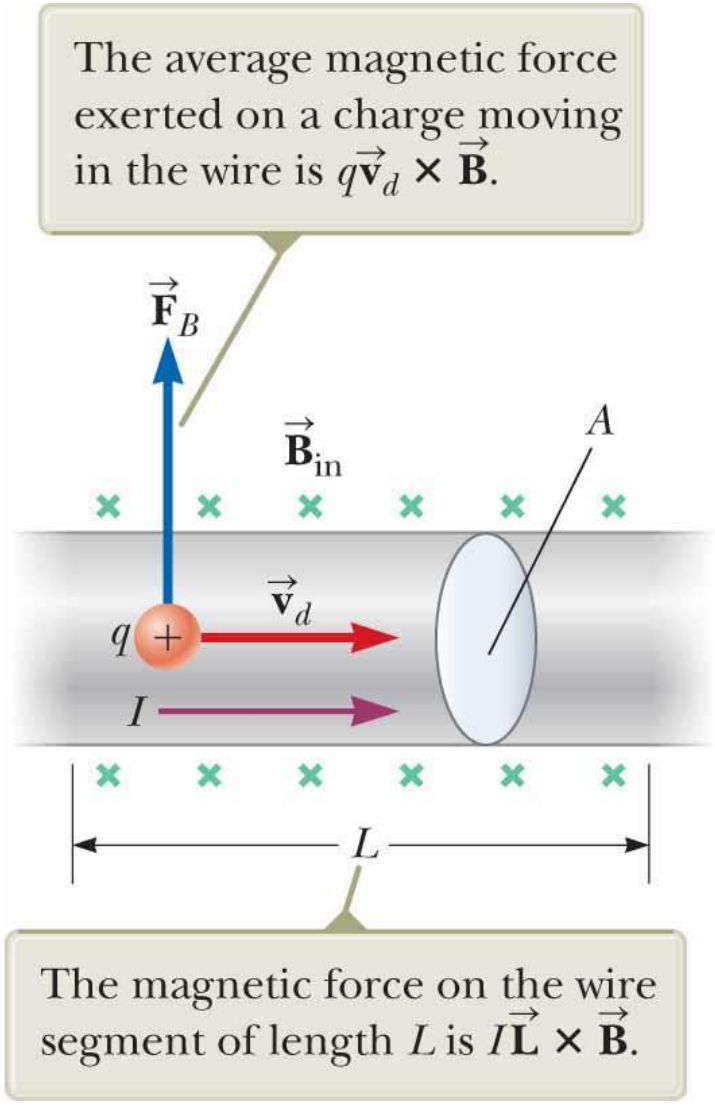
$q\vec{\mathbf{v}}_d \times \vec{\mathbf{B}}$ $\qquad$ $V = AL$

charges in segment: nAL

$$I = nqv_d A$$

$$\vec{\mathbf{F}}_B = \left(q\vec{\mathbf{v}}_d \times \vec{\mathbf{B}}\right) nAL$$

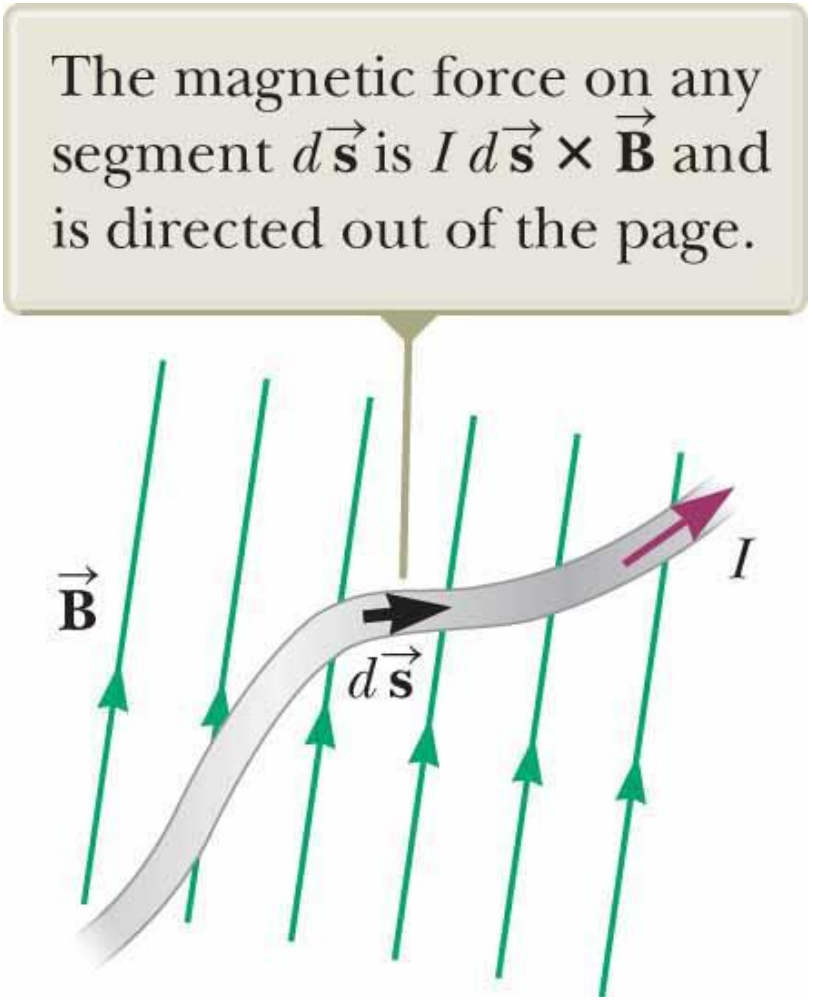
$$\vec{\mathbf{F}}_B = I\vec{\mathbf{L}} \times \vec{\mathbf{B}}$$



Magnetic Force Acting on a Current-Carrying Conductor

$$d\vec{\mathbf{F}}_B = Id\vec{\mathbf{s}} \times \vec{\mathbf{B}}$$

$$\vec{\mathbf{F}}_B = I\int_a^b d\vec{\mathbf{s}} \times \vec{\mathbf{B}}$$



Quick Quiz 28.3

A wire carries current in the plane of this screen toward the top of the screen. The wire experiences a magnetic force toward the right edge of the screen. Is the direction of the magnetic field causing this force

(a) in the plane of the screen and toward the left edge,
(b) in the plane of the screen and toward the bottom edge,
(c) out of the screen, or
(d) into the screen?

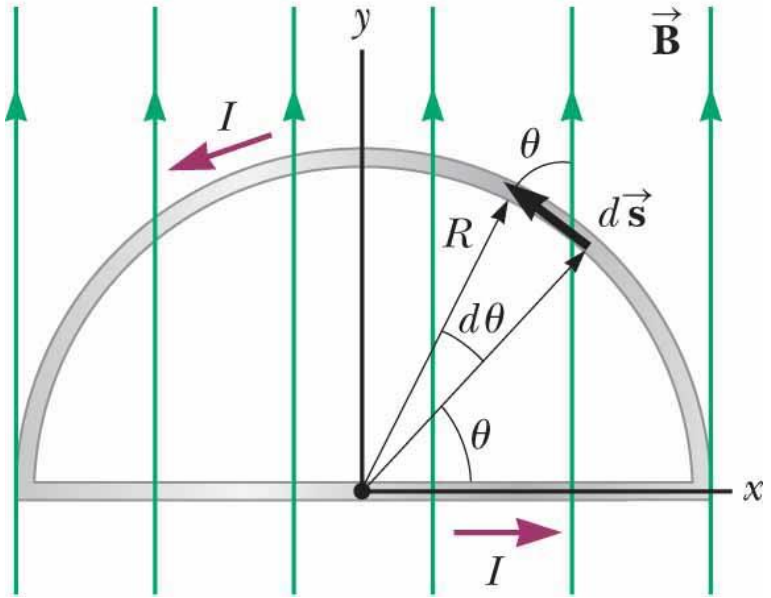
Quick Quiz 28.3

A wire carries current in the plane of this screen toward the top of the screen. The wire experiences a magnetic force toward the right edge of the screen. Is the direction of the magnetic field causing this force

(a) in the plane of the screen and toward the left edge,
(b) in the plane of the screen and toward the bottom edge,
(c) out of the screen, or
(d) into the screen?

Example 28.4:
Force on a Semicircular Conductor

A wire bent into a semicircle of radius R forms a closed circuit and carries a current *I*. The wire lies in the *xy* plane, and a uniform magnetic field is directed along the positive *y* axis as in the figure. Find the magnitude and direction of the magnetic force acting on the straight portion of the wire and on the curved portion.

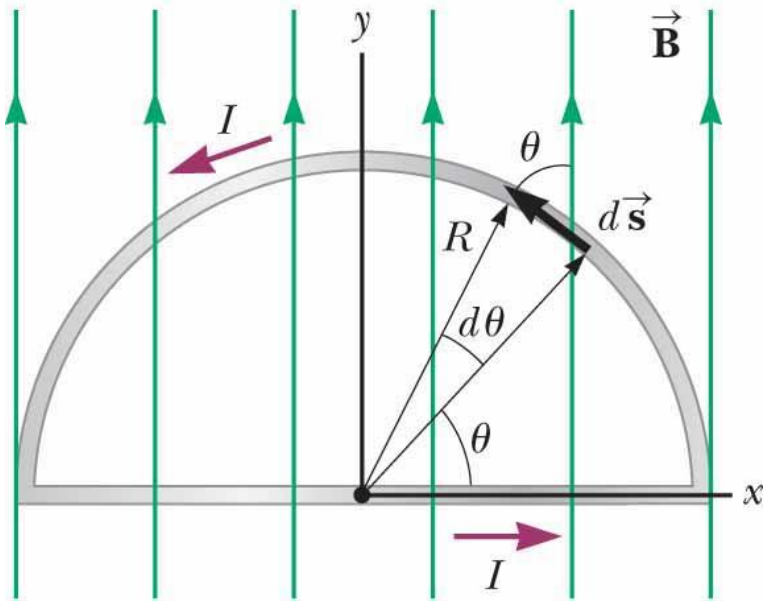


Example 28.4:
Force on a Semicircular Conductor

$$\vec{\mathbf{F}}_1 = I\int_a^b d\vec{\mathbf{s}}\times\vec{\mathbf{B}} = I\int_{-R}^{R} B\,dx\,\hat{\mathbf{k}}$$
$$= 2IRB\,\hat{\mathbf{k}}$$

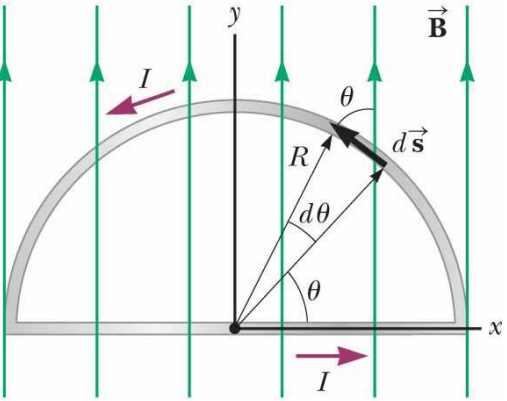
$$d\vec{\mathbf{F}}_2 = Id\vec{\mathbf{s}}\times\vec{\mathbf{B}} = -IB\sin\theta\,ds\,\hat{\mathbf{k}}$$

$$ds = R\,d\theta$$



$$\vec{\mathbf{F}}_2 = -\int_0^{\pi} IRB\sin\theta\,d\theta\,\hat{\mathbf{k}} = -IRB\int_0^{\pi}\sin\theta\,d\theta\,\hat{\mathbf{k}} = -IRB\left[-\cos\theta\right]_0^{\pi}\hat{\mathbf{k}}$$
$$= IRB\left(\cos\pi - \cos 0\right)\hat{\mathbf{k}} = IRB\left(-1-1\right)\hat{\mathbf{k}} = \boxed{-2IRB\hat{\mathbf{k}}}$$

Torque on a Current Loop in a Uniform Magnetic Field



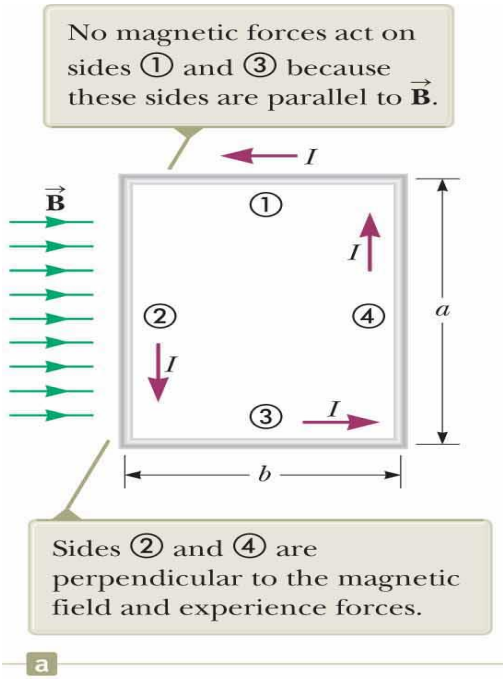
$$F_2 = F_4 = IaB$$

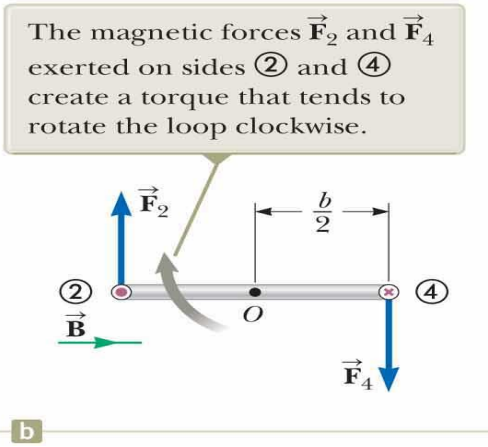
$$\tau_{\max} = F_2 \frac{b}{2} + F_4 \frac{b}{2}$$

$$= (IaB)\frac{b}{2} + (IaB)\frac{b}{2}$$

$$= IabB$$

$$\tau_{\max} = IAB$$



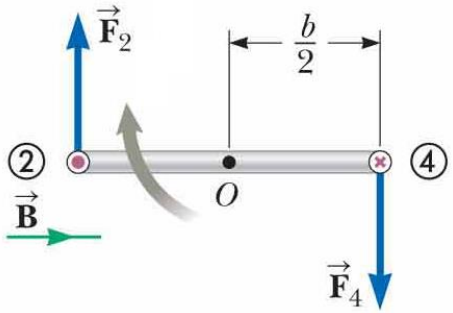


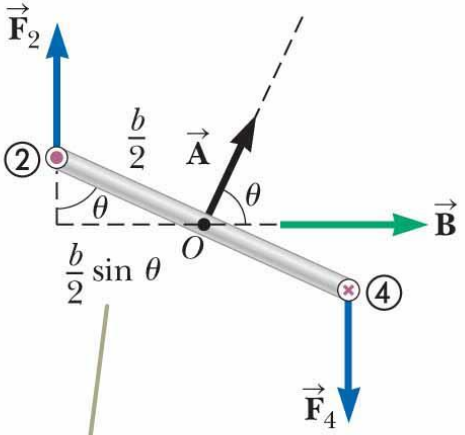
Torque on a Current Loop in a Uniform Magnetic Field

$$\tau = F_2 \frac{b}{2}\sin\theta + F_4 \frac{b}{2}\sin\theta$$

$$= IaB\left(\frac{b}{2}\sin\theta\right) + IaB\left(\frac{b}{2}\sin\theta\right) = IabB\sin\theta$$

$$= IAB\sin\theta$$





When the normal to the loop makes an angle θ with the magnetic field, the moment arm for the torque is $(b/2)\sin\theta$.

Torque on a Current Loop in a Uniform Magnetic Field

$$\vec{\boldsymbol{\tau}} = I\vec{\mathbf{A}} \times \vec{\mathbf{B}}$$

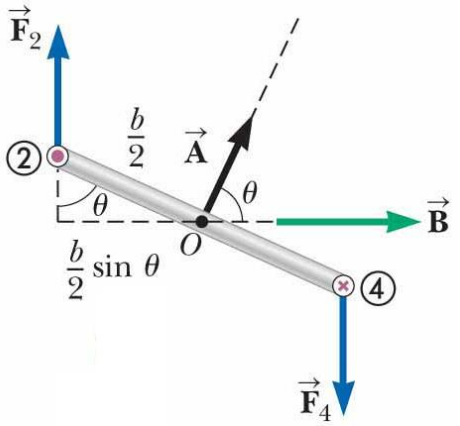
$$\vec{\boldsymbol{\mu}} \equiv I\vec{\mathbf{A}}$$

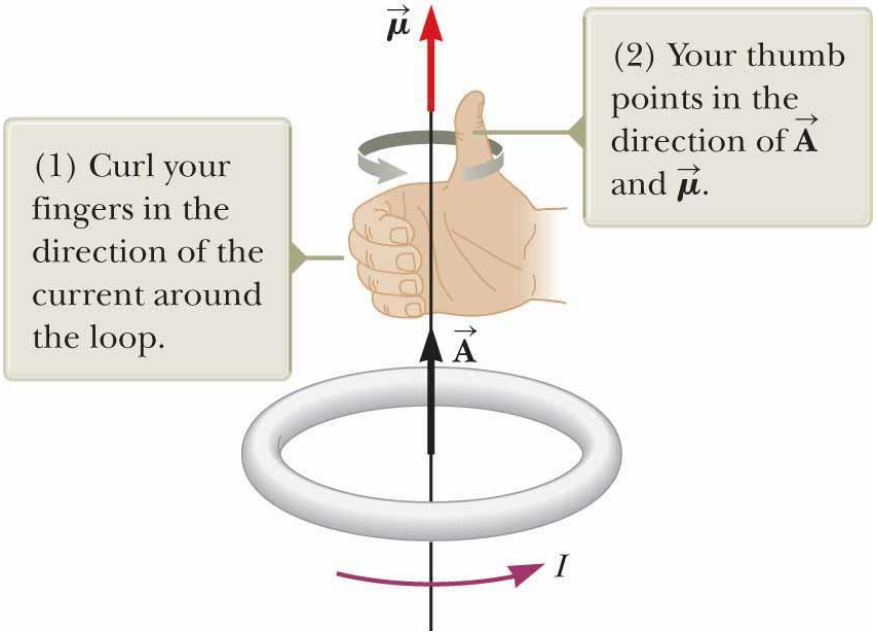
$$\vec{\boldsymbol{\mu}}_{\text{coil}} = NI\vec{\mathbf{A}}$$

$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{\mu}} \times \vec{\mathbf{B}}$$

$$\vec{\boldsymbol{\tau}} = \vec{\mathbf{p}} \times \vec{\mathbf{E}}$$

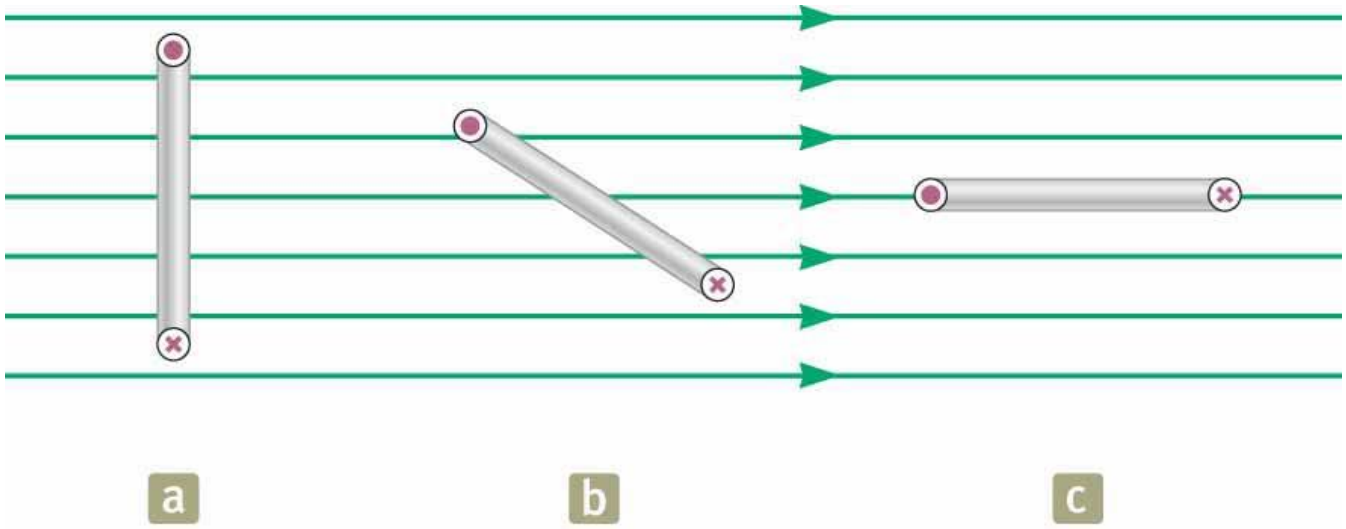
$$U_B = -\vec{\boldsymbol{\mu}} \cdot \vec{\mathbf{B}}$$





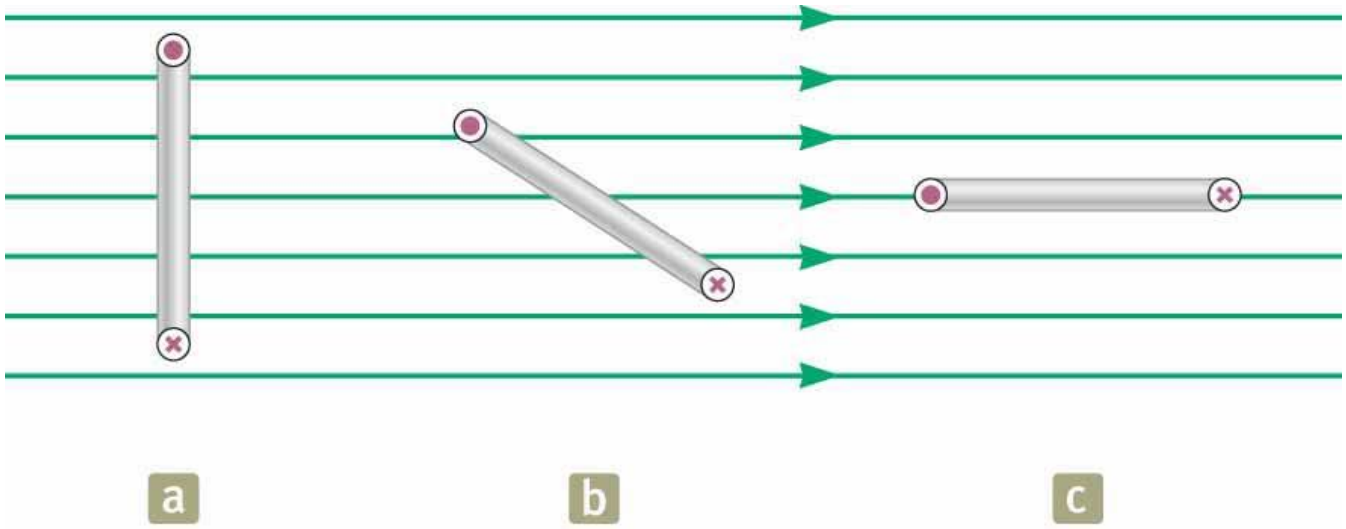
Quick Quiz 28.4 Part I

Rank the magnitudes of the torques acting on the rectangular loops (a), (b), and (c) shown edge-on in the figure from highest to lowest. All loops are identical and carry the same current.



Quick Quiz 28.4 Part I

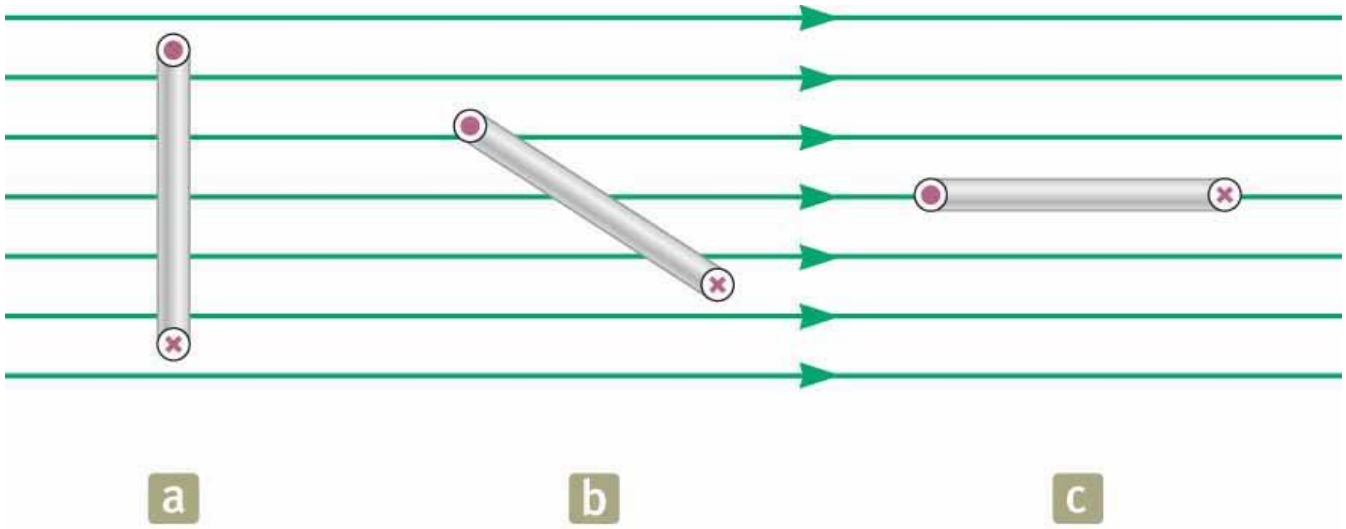
Rank the magnitudes of the torques acting on the rectangular loops (a), (b), and (c) shown edge-on in the figure from highest to lowest. All loops are identical and carry the same current.



(c), (b), (a)

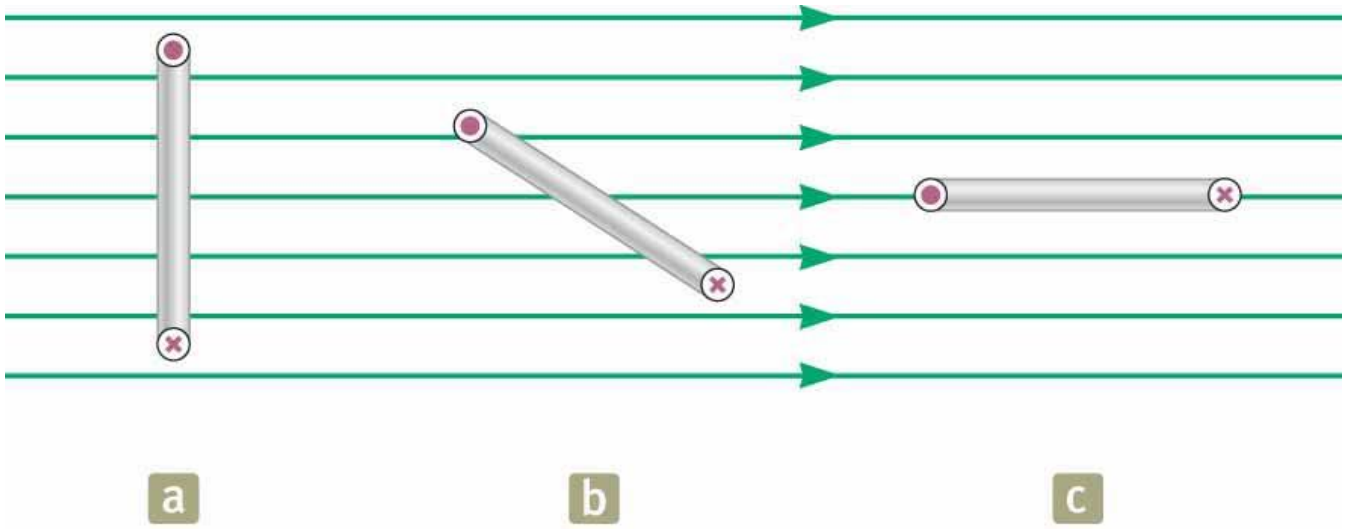
Quick Quiz 28.4 Part II

Rank the magnitudes of the net forces acting on the rectangular loops shown in the figure from highest to lowest.



Quick Quiz 28.4 Part II

Rank the magnitudes of the net forces acting on the rectangular loops shown in the figure from highest to lowest.



(a) = (b) = (c)

Example 29.5 The Magnetic Dipole Moment of a Coil

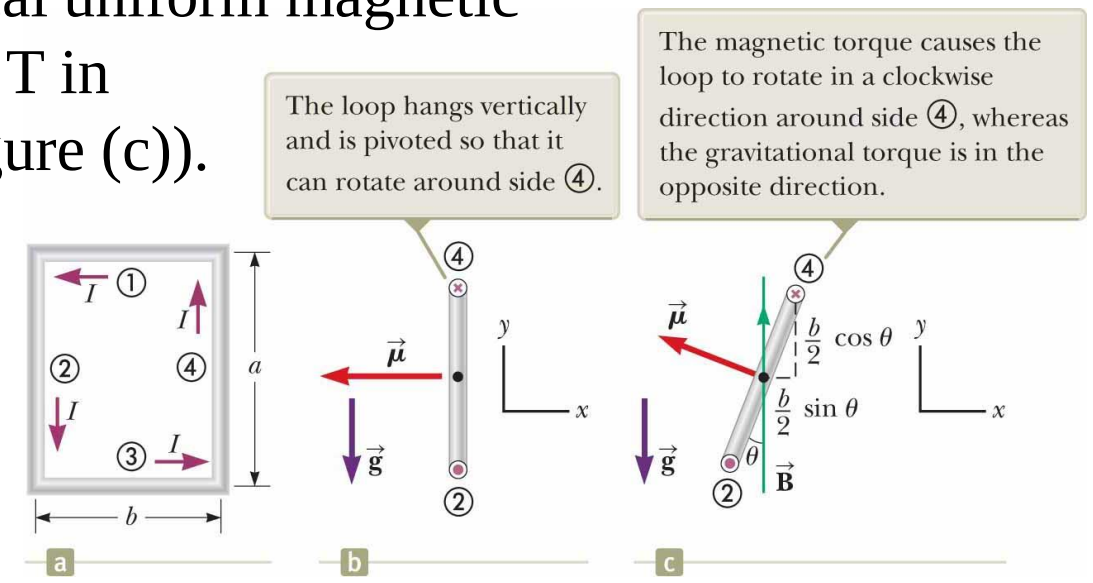
A rectangular coil of dimensions 5.40 cm × 8.50 cm consists of 25 turns of wire and carries a current of 15.0 mA. A 0.350-T magnetic field is applied parallel to the plane of the coil.

(A) Calculate the magnitude of the magnetic dipole moment of the coil.

(B) What is the magnitude of the torque acting on the loop?

Example 28.5: Rotating a Coil

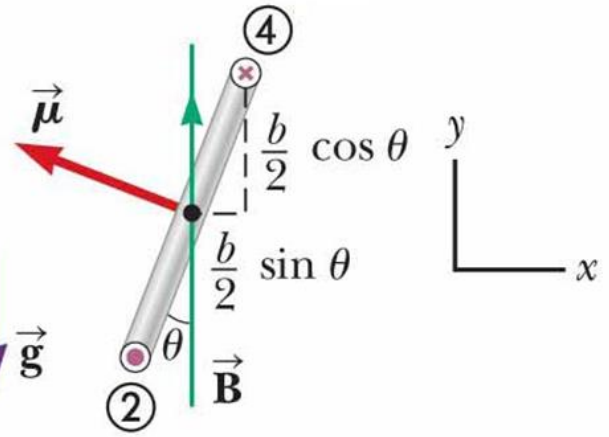
Consider the loop of wire in figure (a). Imagine it is pivoted along side 4, which is parallel to the *z* axis and fastened so that side 4 remains fixed and the rest of the loop hangs vertically in the gravitational field of the Earth but can rotate around side 4 (figure (b)). The mass of the loop is 50.0 g, and the sides are of lengths a = 0.200 m and b = 0.100 m. The loop carries a current of 3.50 A and is immersed in a vertical uniform magnetic field of magnitude 0.010 0 T in the positive *y* direction (figure (c)). What angle does the plane of the loop make with the vertical?



Example 28.5: Rotating a Coil

$$\vec{\tau}_B = \vec{\boldsymbol{\mu}} \times \vec{\mathbf{B}} = -\mu B \sin\left(90° - \theta\right)\hat{\mathbf{k}}$$
$$= -IAB\cos\theta\,\hat{\mathbf{k}} = -IabB\cos\theta\,\hat{\mathbf{k}}$$

$$\vec{\tau}_g = \vec{\mathbf{r}} \times m\vec{\mathbf{g}} = mg\frac{b}{2}\sin\theta\,\hat{\mathbf{k}}$$



$$\sum \vec{\tau} = -IabB\cos\theta\,\hat{\mathbf{k}} + mg\frac{b}{2}\sin\theta\,\hat{\mathbf{k}} = 0$$

$$IabB\cos\theta = mg\frac{b}{2}\sin\theta \Rightarrow \tan\theta = \frac{2IaB}{mg}$$

$$\theta = \tan^{-1}\left[\frac{2\left(3.50\text{ A}\right)\left(0.200\text{ m}\right)\left(0.0100\text{ T}\right)}{\left(0.0500\text{ kg}\right)\left(9.80\text{ m/s}^2\right)}\right] = \boxed{1.64°}$$

The Hall Effect

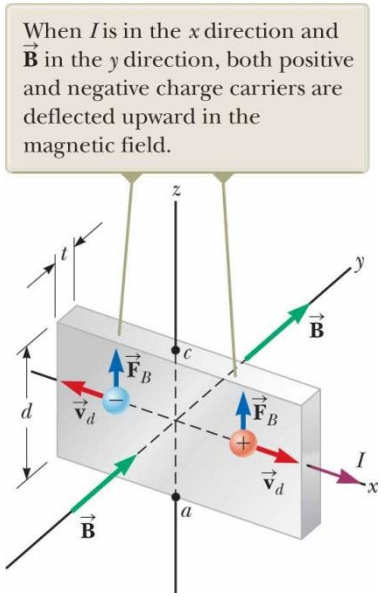
$$qv_d B = qE_{\mathrm{H}} \Rightarrow E_H = v_d B$$

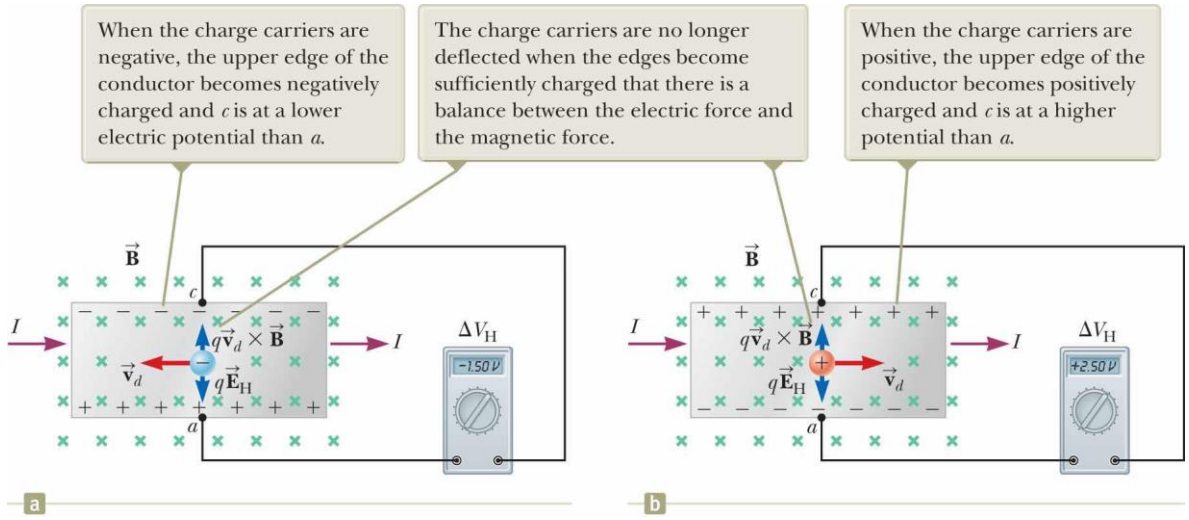
$$\Delta V_{\mathrm{H}} = E_{\mathrm{H}} d = v_d B d$$

$$v_d = \frac{I}{nqA} \qquad \Delta V_{\mathrm{H}} = \frac{IBd}{nqA}$$

$$\Delta V_{\mathrm{H}} = \frac{IB}{nqt}$$

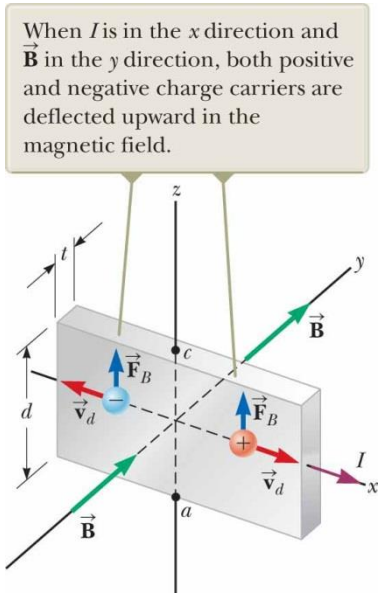
$$= \frac{R_{\mathrm{H}} IB}{t}$$

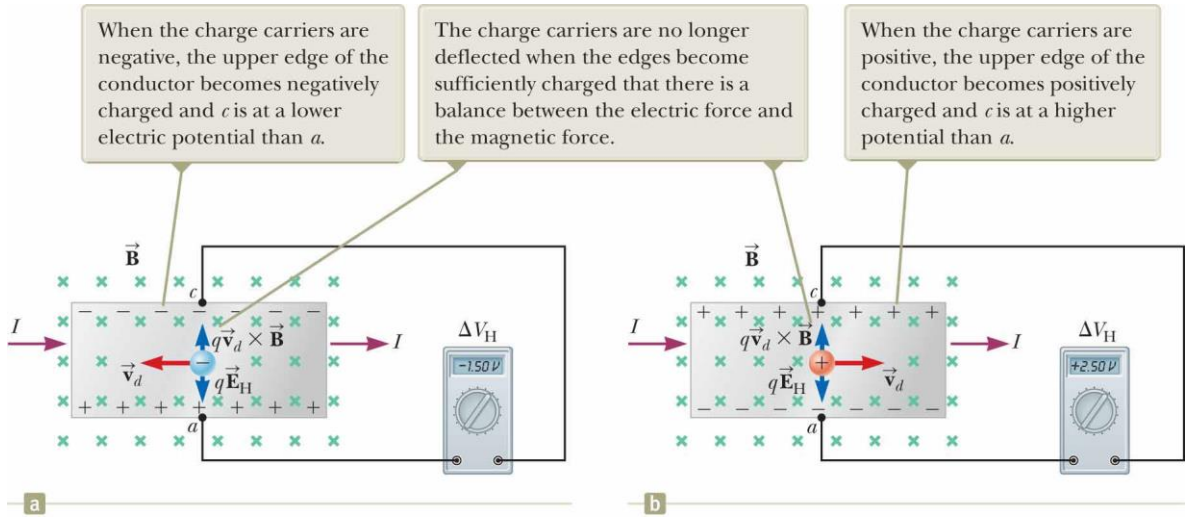




Example 28.6:
The Hall Effect for Copper

A rectangular copper strip 1.5 cm wide and 0.10 cm thick carries a current of 5.0 A. Find the Hall voltage for a 1.2-T magnetic field applied in a direction perpendicular to the strip.

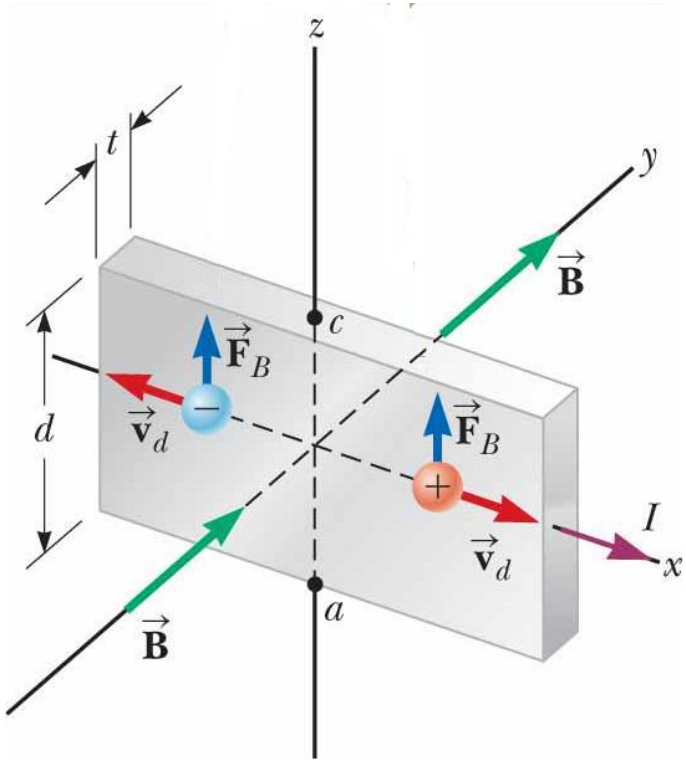




Example 28.6: The Hall Effect for Copper

$$n = \frac{N_A}{V} = \frac{N_A \rho}{M}$$

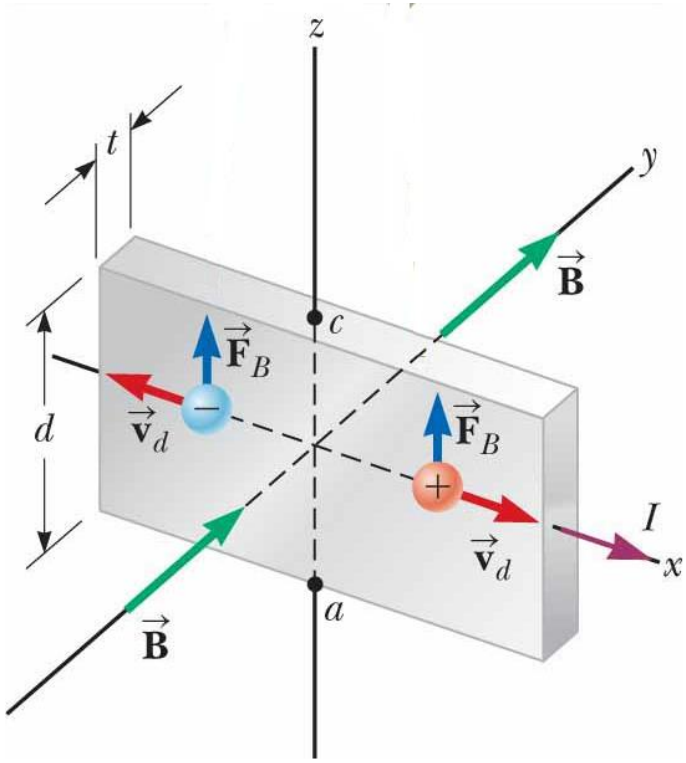
$$\Delta V_{\mathrm{H}} = \frac{IB}{nqt} = \frac{MIB}{N_A \rho q t}$$



$$\Delta V_{\mathrm{H}} = \frac{(0.0635\ \mathrm{kg/mol})(5.0\ \mathrm{A})(1.2\ \mathrm{T})}{(6.02\times10^{23}\ \mathrm{mol^{-1}})(8920\ \mathrm{kg/m^3})(1.60\times10^{-19}\ \mathrm{C})(0.0010\ \mathrm{m})}$$
$$= 0.44\ \mu\mathrm{V}$$

Example 28.6:
The Hall Effect for Copper

What if the strip has the same dimensions but is made of a semiconductor? Will the Hall voltage be smaller or larger?



Assessing to Learn

In a certain region of space there is a uniform magnetic field pointing in the positive z direction ($+z$). In what direction should a negative point charge move to experience a force in the positive x direction ($+x$)?

1. in the positive z direction ($+z$)
2. in the negative z direction ($-z$)
3. in the positive x direction ($+x$)
4. in the negative x direction ($-x$)
5. in the positive y direction ($+y$)
6. in the negative y direction ($-y$)
7. It can move in any direction.
8. It is impossible for the force to be in the $+x$ direction when the magnetic field is in the $+z$ direction.

Assessing to Learn

In each of the following situations, point charge *q* moves in a uniform magnetic field B. The strength of the magnetic field is indicated by the density of field lines. In each situation, the initial speed *v* of the charge is the same. For which situation(s) will the charge *q* travel the longest distance in a certain time *T*?

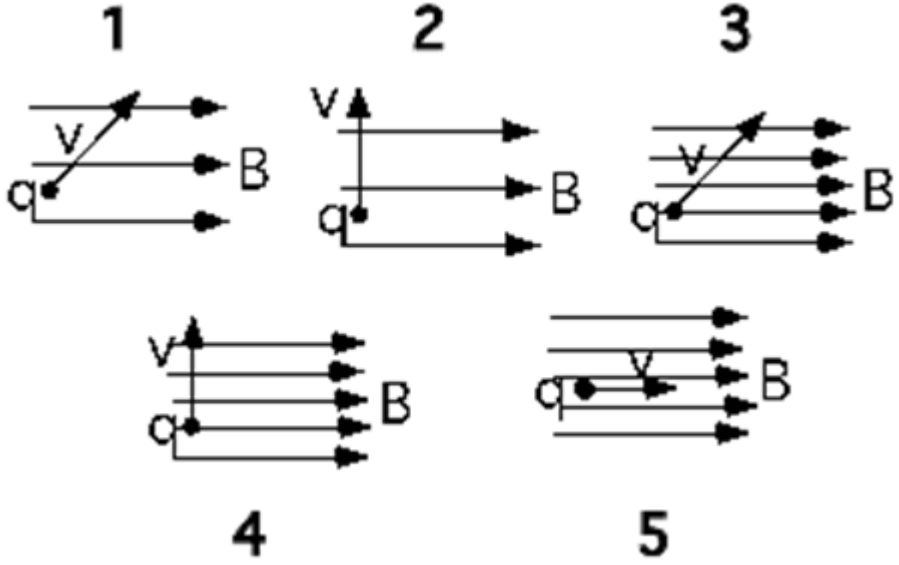
1. 1 2. 2 3. 3

4. 4 5. 5 6. 1 & 3

7. 2 & 4 8. 1, 2, 3 & 4

9. 1, 2, 3, 4 & 5

10. Cannot be determined



Assessing to Learn

In each of the following situations, point charge *q* moves in a uniform magnetic field B. The strength of the magnetic field is indicated by the density of field lines. In each situation, the initial speed *v* of the charge is the same. For which situation(s) will the charge *q* have the largest displacement in a certain time *T*?

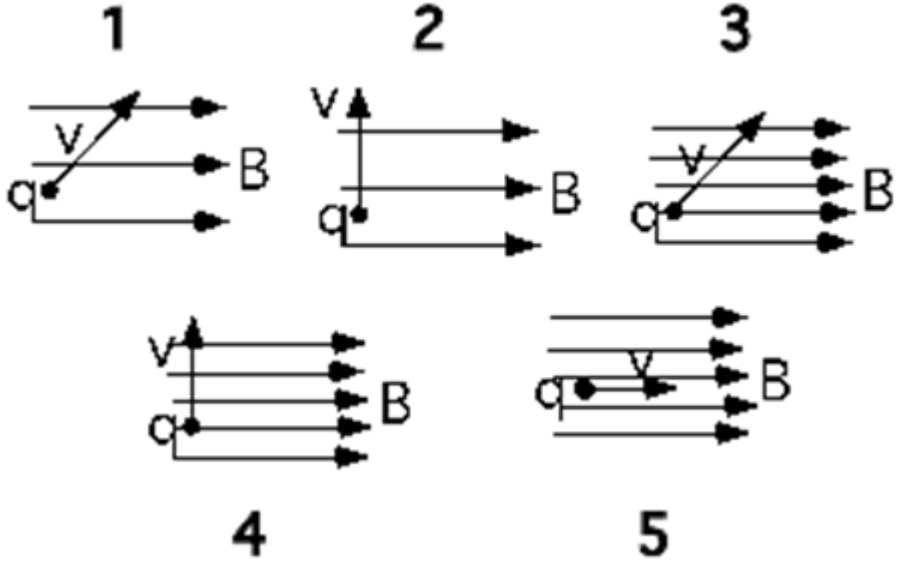
1. 1 2. 2 3. 3

4. 4 5. 5 6. 1 & 3

7. 2 & 4 8. 1, 2, 3, 4 & 5

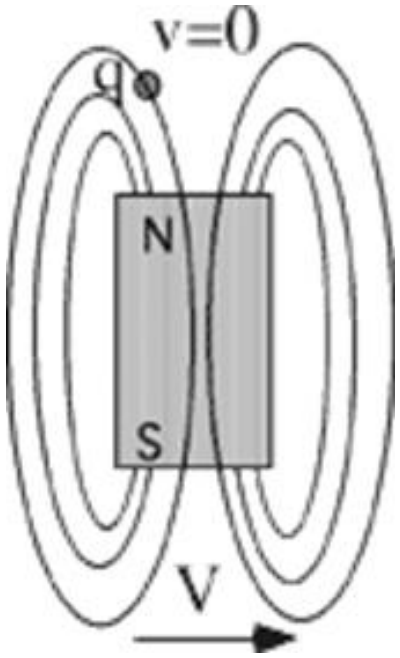
9. None of the above

10. Cannot be determined



Assessing to Learn

A bar magnet moving with speed *V* passes below a stationary charge *q*. What can be said about the magnitude of the magnetic forces on the bar magnet (F_b) and on the charge *q* (F_q).

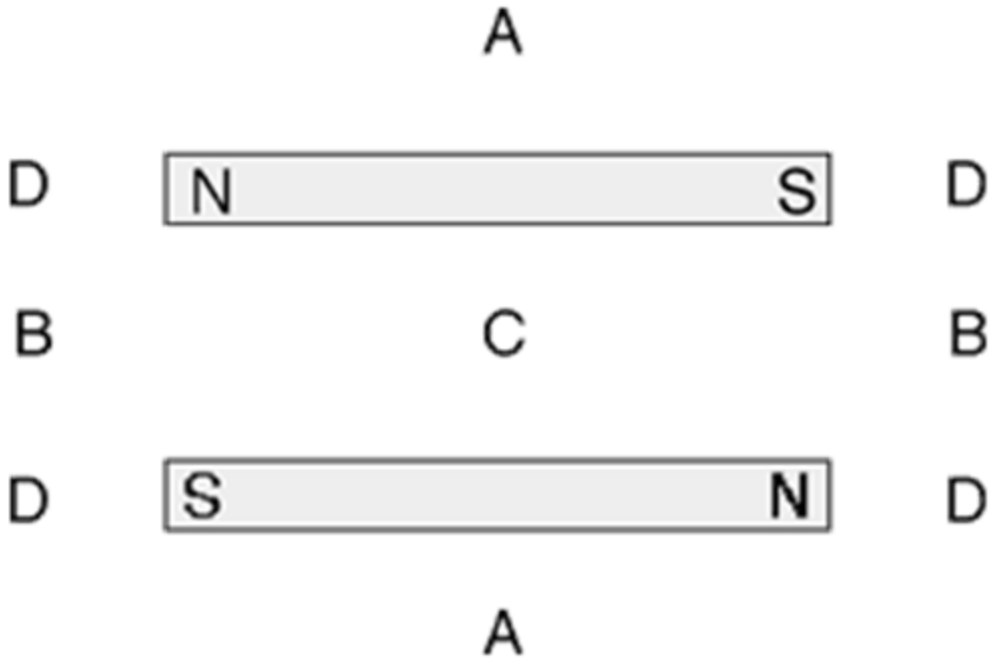


1. F_{bar} and F_q are both zero.
2. F_{bar} is zero and F_q is **not** zero.
3. F_{bar} is **not** zero and F_q is zero.
4. F_{bar} and F_q are both **non-**zero.

Assessing to Learn

Two identical bar magnets are placed rigidly and anti-parallel to each other as shown. At what locations, if any, is the net magnetic field close to zero?

1. A only
2. B only
3. C only
4. D only
5. A and B
6. A, B, and C
7. C and D
8. None of the above.



Assessing to Learn

A charged particle moves into a region containing both an electric field and a magnetic field. Which of the statements below is/are true?

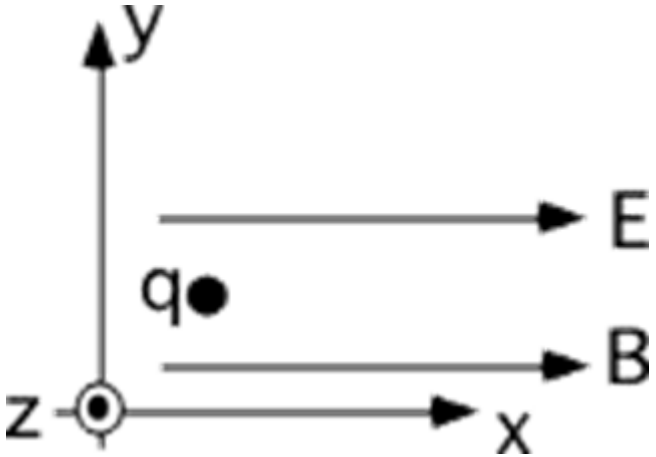
A. The particle cannot accelerate in the direction of B.
B. The path of the particle must be a circle.
C. Any change in the particle's kinetic energy is caused by the E field.

1. Only A
2. Only B
3. Only C
4. Both A and B
5. Both A and C
6. Both B and C
7. All are true.
8. None are true.

Assessing to Learn

A charge is released from rest in E and B fields. Both fields point along the *x* axis. Which of the following statements regarding the charge's motion are correct?

1. The charge will travel along a straight-line path.
2. The charge's speed will change as it travels.
3. The charge will travel in a helical path.
4. The charge will travel in a helical path of increasing pitch.
5. The charge will travel in a circle in the x-y plane.
6. 1 and 2 only
7. 2 and 4 only
8. None of the above



Assessing to Learn

A charge has an initial velocity parallel to the y-axis in E and B fields. Both fields point along the *x* axis. Which of the following statements regarding the charge's motion are correct?

1. The charge will travel along a straight-line path.
2. The charge's speed will change as it travels.
3. The charge will travel in a helical path.
4. The charge will travel in a helical path of increasing pitch.
5. The charge will travel in a circle in the *x*-*y* plane.
6. 1 and 2 only
7. 2 and 4 only
8. None of the above

